

Review article

Transforming education with AI: A systematic review of ChatGPT's role in learning, academic practices, and institutional adoption

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ABSTRACT

The integration of AI tools like ChatGPT into education has generated significant interest due to their potential to transform learning environments by providing personalized learning, automating tasks, and improving student engagement. However, gaps remain in the literature, particularly in comparing AI-supported education methods with traditional approaches and understanding ChatGPT's specific role in academic writing, literature reviews, and teacher development. This study addresses these gaps through a systematic literature review (SLR), evaluating the effectiveness of AI tools versus traditional teaching approaches. It focuses on ChatGPT's application in areas such as academic writing, lesson planning, student assessment, and the professional development of educators. Additionally, the study explores institutional strategies for balancing the benefits of AI with the need to maintain academic integrity and educational quality. The methodology involved a systematic search across academic databases with a structured analysis of key studies in AI-supported education. The main contributions include identifying the comparative advantages and limitations of AI in enhancing student learning, offering best practices for incorporating AI tools into teaching, and examining prompt engineering as a crucial factor in optimizing AI usage. The study reveals that ChatGPT and other AI tools significantly enhance educational efficiency and engagement, but they require careful management to prevent over-reliance. The study advocates for a balanced approach, integrating both AI and traditional methods to achieve optimal educational outcomes while maintaining academic integrity.

1. Introduction

Over the past decade, Artificial Intelligence (AI) has significantly transformed education, redefining both learning methodologies and administrative processes [1]. As educators and institutions increasingly adopt AI technologies, evaluating their effectiveness compared to traditional teaching approaches has emerged as a critical area of inquiry [2]. Among these advancements, ChatGPT, developed by OpenAI, stands out for its contributions to academic writing, streamlined literature reviews, and professional development for educators. The Generative Pre-trained Transformer (GPT) series that include GPT-1 through GPT-4 and their theoretical successor GPT-4o [3,4] have

demonstrated their ability to simplify complex information processing, providing students with coherent, contextually relevant content while reducing cognitive and temporal demands [5]. The COVID-19 pandemic further accelerated the adoption of AI technologies in education, as institutions turned to digital solutions to sustain learning during periods of confinement [6]. Additionally, organizations such as UNESCOs have advocated for inclusive policies to ensure that AI-driven educational benefits are accessible to all students, regardless of socioeconomic status [7]. AI tools like ChatGPT offer transformative potential in education by enabling formative assessments, personalized learning experiences, and access to vast amounts of information and perspectives not readily available in traditional classrooms.

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Despite its potential, research on the transformative impact of ChatGPT and similar AI tools in education remains limited and fragmented. Earlier studies [18,44] have highlighted advantages such as accessibility and efficiency but often overlooked potential drawbacks, including over-reliance, critical thinking challenges, and limitations in fostering deeper student engagement. Other researchers [42,43,45] have compared traditional and AI-supported teaching methods but neglected broader applications of AI in areas such as data analysis, teaching materials development, language learning, and plagiarism prevention.

ChatGPT's effectiveness in academic writing, literature review synthesis, and translation has been acknowledged [9,10]. However, studies [14,60,62,90] have not thoroughly explored its writing efficacy, particularly in producing cohesive and high-quality literature reviews. Furthermore, as higher education institutions increasingly adopt AI tools, they face significant challenges in balancing opportunities with risks, including academic integrity concerns and potential over-dependence on AI technologies. While some studies [19,54,58] have examined ChatGPT's impact on academic integrity, they have not sufficiently addressed how institutions worldwide manage these challenges while maximizing AI's benefits.

Prompt engineering, a key practice for optimizing AI tools for specific educational needs, is gaining prominence, especially in teaching preparation and assessment [13,14]. While prior research [47–49] has explored ChatGPT's role in lesson planning and fostering critical thinking, gaps remain in understanding how AI can comprehensively enhance teacher preparation and student assessments. Additionally, broader studies [8,16] addressing the opportunities and challenges of AI often neglect the technical details required to create effective prompts for maximizing AI's potential in career advancement.

This study addresses these critical gaps by thoroughly investigating the role of ChatGPT and other AI tools in education. It evaluates their effectiveness in improving academic writing, literature review synthesis, lesson planning, student assessments, and professional development for educators. Furthermore, it analyzes how institutions integrate AI tools into ERP and e-learning systems responsibly through policy development and strategies for balancing innovation with academic integrity. By offering insights into AI's role in key areas such as data analysis, teaching materials development, language learning, and plagiarism prevention [54], this study provides a balanced perspective on the strengths and limitations of AI-driven methods compared to traditional approaches. Table 1 shows the gaps that have been covered by this study.

Eventually, the scope of this study is to thoroughly investigate the role of ChatGPT and other AI tools in education, highlighting their effectiveness in improving academic writing, literature reviews, lesson planning, student assessments, educator professional development, and prompt engineering. In this context, the use of ChatGPT and other AI technologies opens up opportunities for students to access a vast amount of information and perspectives that may not be readily available in traditional classrooms (Fig. 1)

1.1. Motivations

The integration of AI tools like ChatGPT into educational settings presents several key challenges and opportunities [7–9]. A systematic assessment is required to evaluate how AI-supported approaches compare to traditional teaching strategies in terms of enhancing student engagement, fostering critical thinking, and improving academic performance. It is also essential to assess their effectiveness in achieving educational objectives [10]. Additionally, concerns about the impact of ChatGPT on the quality and originality of academic outputs, particularly in academic writing, text translation, and literature reviews, necessitate further examination [11]. There is a noticeable gap in the literature regarding how educational institutions globally are managing the integration of AI technologies while upholding academic standards [12]. This highlights the crucial need for research on the balance between

Table 1
literature reviews related work.

Contribution Points	[18, 44]	[42, 43, 45]	[9, 10]	[19, 54, 58]	[47–49]	This Study
Comparison of AI-supported vs. traditional teaching methods	✓	◆	◆	◆	◆	✓
Examination of ChatGPT's role in academic writing, literature reviews, and translation	◆	✓	✓	◆	◆	✓
Exploring strategies for institutional AI integration with a focus on policy development for balancing AI and academic integrity	◆	✓	◆	✓	◆	✓
Practical applications of ChatGPT for professional development (lesson planning, assessments, personalized feedback)	◆	✓	✓	◆	✓	✓
Optimal strategies for prompt engineering to align with educational goals (e.g., evaluations, focused feedback)	◆	◆	✓	◆	✓	✓
Insights into AI's role in areas like data analysis, teaching material development, language learning, and plagiarism prevention	◆	◆	◆	◆	◆	✓

using AI tools and maintaining educational quality and integrity. Furthermore, the professional development of educators represents a vital area where AI technologies can enhance classroom preparation and assessment methods. Given the nascent nature of ChatGPT's application in education and its varied effects, there is a pressing need for systematic LR research. Motivated by the potential of AI and ChatGPT to transform education, this study seeks to critically evaluate their efficacy, ethical considerations, and develop best practices for their integration.

1.2. Research contributions

Although the studies [10,13,14] have discussed the need for ChatGPT and the significant features of AI technologies in the education system, this survey extends the discussion to a broader range of topics, including AI in education, the comparison between AI-supported and traditional teaching methods, ChatGPT's effectiveness in academic writing, and the responses of global educational institutions to AI. This breadth provides an overview of the state of AI in education, which is valuable for readers and researchers looking to understand AI's impact on the education sector. The primary contributions of this paper may be briefly stated as follows:

- This research provides a detailed comparison of AI-supported educational methods, particularly those involving ChatGPT, against traditional teaching practices. It identifies specific contexts where AI tools excel, while also highlighting areas where traditional methods remain essential.
- By examining ChatGPT's application in academic writing, this study offers insights into its role of drafting, structuring, and refining academic texts. It further sheds light on ChatGPT's capabilities in conducting preliminary in academic writing, literature reviews, and translation.
- This work explores strategies employed by global higher education institutions for integrating AI tools, focusing on policy development aimed at balancing innovation with academic integrity. It presents

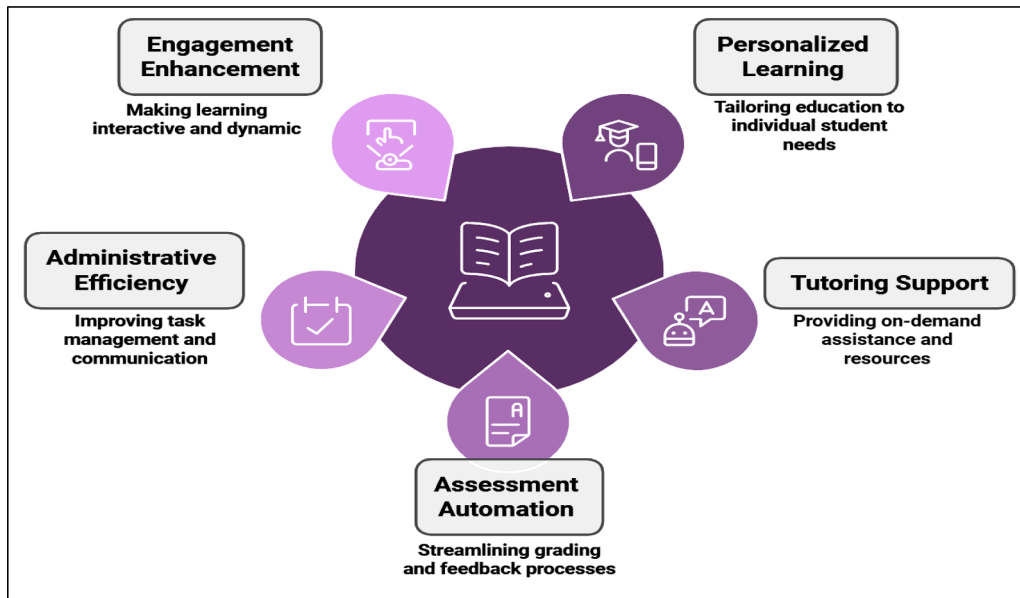


Fig. 1. ChatGPT and other AI applications in the education domain.

- an overview of diverse institutional practices, ranging from strict regulations to adaptive guidelines that promote responsible AI use.
- The study outlines practical applications of ChatGPT for professional development, including lesson planning, assessment design, and personalized feedback.
 - This study outlines optimal strategies for formulating prompts that fit with instructional objectives, allowing educators to fully leverage ChatGPT's capabilities in activities such as customized evaluations and focused feedback, hence facilitating various educational aims.

1.3. Organization of the paper

This paper is structured in six sections to provide a comprehensive exploration of the integration of AI technologies, and ChatGPT, within educational contexts. It begins with an introduction that outlines the research topic, motivations, and contributions, followed by section two with a detailed literature review and conceptual framework. The research methodology in section three then describes the systematic literature review (SLR) process, including the search strategy, inclusion and exclusion criteria, and methods for data extraction and analysis. The findings are presented next in section four, organized according to the four research questions that guide the study. This is followed by a

discussion in section five. Finally, section six of the paper concludes with a study limitation, a summary of key insights and suggests potential directions for future research. Fig. 2 below depicts the research structure.

2. Literature review and conceptual framework

This section shows the popularity of ChatGPT in education; the most recent excitement about ChatGPT; and the evolution that has taken place so far with regard to tools based on AI for teaching and learning. It explains emerging methodologies, such as Generative AI and Prompt Engineering, and discusses the integration of AI technologies into the reinforcement of educational practices and outcomes.

2.1. The recent popularity of ChatGPT in education

This review focused on ChatGPT-4 which has been reported to handle more complex tasks with greater efficiency and can process a wider range of prompts with improved coherence [15]. In comparison with GPT 3, ChatGPT-4 has shown promise in assisting with academic writing by producing content that is both cohesive and contextual which can help with literature reviews by creating summaries of existing

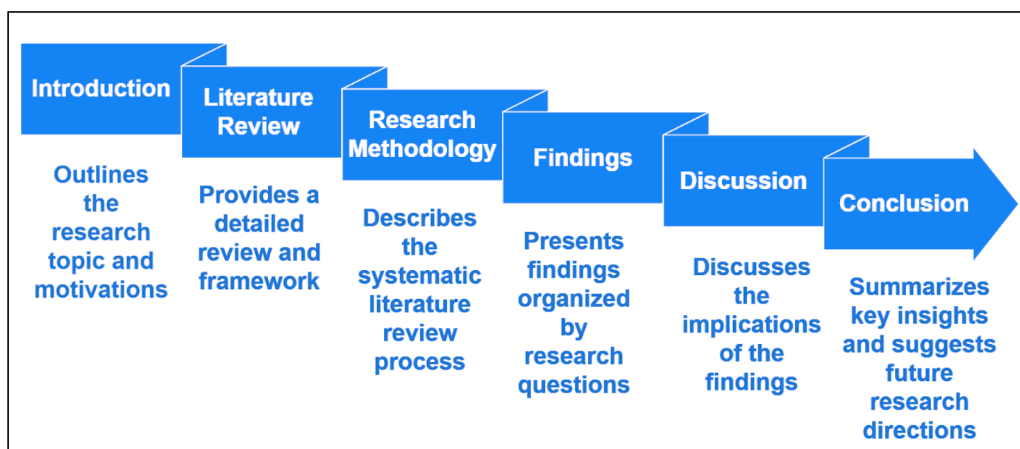


Fig. 2. The research structure.

research, highlighting significant themes, composing sections of research articles, and even recommending possible research needs [11]. However, the literature notes that ChatGPT's efficiency is restricted by its training data, which may not cover the most recent studies [16]. The implications of ChatGPT in the field of education were further explored in a review by Mhlanga [17]. He conducted a careful analysis of eight articles that delved into the subject matter of ChatGPT and explored the implications of integrating ChatGPT in education. It discusses ChatGPT's potential to personalize learning, provide 24/7 accessibility, facilitate language learning, streamline assessment processes, aid in content generation and curriculum development, support teachers' professional development, and enhance accessibility and inclusion. The authors of [9,18] discussed the use of ChatGPT in education, focusing on its potential, challenges, and strategies to address issues such as accuracy, reliability, and plagiarism. It discusses how ChatGPT can be used to enhance teaching and learning, as well as potential issues associated with its use, such as generating incorrect information and facilitating plagiarism. The recent popularity of ChatGPT in education reflects its growing influence and potential as a transformative tool. Its adoption is driven by its ability to assist in various educational activities, such as lesson planning, grading, and providing personalized feedback. This surge in interest highlights the need to explore how ChatGPT can be effectively integrated into teaching practices and its impact on both educators and students.

2.2. Evolution of AI tools in education

The evolution of AI in education demonstrates a continuous progression, originating from rule-based systems and intelligent tutoring in the 1990s, advancing via adaptive learning technologies and natural language processing (NLP) applications [19]. Initially, educational AI systems were designed to facilitate personalized learning and automate grading, enabling educators to dedicate more time to interactive learning experiences. As time progressed, advanced machine learning algorithms facilitated predictive analytics, allowing AI systems to discern student learning patterns, forecast performance, and provide customized interventions. The emergence of generative AI models, like ChatGPT, signified a significant advancement [20]. In contrast to previous AI, generative models can autonomously produce human-like replies, rendering them exceptionally versatile for diverse educational activities, including individualized feedback and the creation of interactive lesson content. These tools can facilitate real-time interaction with students, assist in creative work, and offer accessible assistance beyond conventional classroom hours. This evolution has resulted in a present situation where generative AI may support instructors and serve as a co-creator of educational content, hence boosting accessibility, adaptability, and engagement in teaching and learning [21].

2.3. Generative AI (GAI) and prompt engineering

Unlike traditional AI, GAI is designed to produce novel outputs that demonstrate human creative abilities, generating new content text, images, and music, producing coherent and context-specific text responses or even code based on user input, similar to ChatGPT [22]. This technology aims to develop systems that can grasp, learn, and tackle any intellectual task that humans can. To achieve this, prompt engineering is crucial, focusing on designing and implementing inputs that elicit desired behaviors and outputs from AI models, resolving the challenge of aligning AI goals with human intentions [23]. Prompt engineering summarizes a multidisciplinary process guided by insights and practices borrowed from cognitive psychology, linguistics, and computer science in the design, construction, and fine-tuning of prompts for application in Human-Computer Interaction (HCI) [24], especially under applications of Natural Processing Language (NLP) what drives improvements in conversational agent efficiency, mostly based on chatbots, virtual assistants, and language models [25]. The primary objective of prompt

engineering is to unlock the potential of AI systems to produce highly accurate, relevant, and coherent responses, ultimately delivering a seamless and satisfying user experience [26].

2.4. Integration of AI technology in the field of education

Integration in education refers to the systematic amalgamation of diverse subjects, programs, or pedagogical approaches to create a cohesive and comprehensive educational experience [27]. This process entails the synthesis of distinct disciplines, including mathematics, science, and language arts, to foster a profound comprehension of the interrelationships among them [28]. Proficient integration promotes the development of critical thinking, problem-solving, and collaborative competencies, thereby yielding more engaging and significant educational outcomes. AI technologies, including Intelligent Tutoring Systems (ITS) and adaptive learning platforms, possess the capability to customize educational materials to align with the distinct needs of individual learners [29]. Empirical research has demonstrated that these technological advancements can facilitate enhanced educational outcomes by addressing the specific lacks of students and delivering targeted feedback [30,31]. For example, research conducted by [32] revealed that AI-driven adaptive learning systems substantially elevated student performance in large-scale online courses when contrasted with conventional instructional methods. Traditional educational approaches, characterized by an emphasis on human interaction and experiential learning, present advantages in domains where AI systems may exhibit limitations. In fields requiring advanced critical analysis and innovation, traditional teaching methods demonstrate greater effectiveness. Nonetheless, current research supports a hybrid teaching model combining AI-enhanced techniques with conventional instructional strategies for improved student outcomes [33].

3. Methodology

3.1. SLR

This study followed the Systematic Literature Review (SLR) approach suggested by Kitchenham [34]. To maintain consistency, this study employed standardized data extraction forms derived from Kitchenham's technique [35,36]. Through the systematization of our data collection, critical themes, insights, and trends concerning AI applications in education have been identified, and a comprehensive comparative analysis of AI-supported and traditional methodologies have been performed. It is essential to acknowledge that recognizing the necessity for a review holds equal significance to the review process itself (Fig. 3). Based on findings from existing research on the integration of AI and ChatGPT into educational settings, this systematic review examines the impact of these technologies on educational effectiveness, focusing specifically on student engagement, academic performance, educator development, and institutional strategies for balancing opportunities and risks. The review addresses the following research

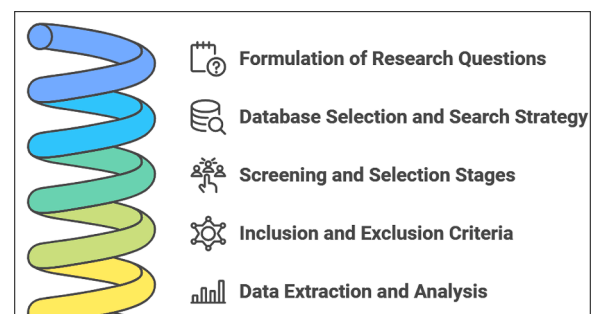


Fig. 3. The systematic review procedure.

questions, as detailed in Table 2:

RQ1: What is the comparative effectiveness of AI-supported education methods, versus traditional teaching methods in enhancing student academic performance considering their respective advantages and disadvantages?

RQ2: How does ChatGPT demonstrate its capabilities and effectiveness in academic writing and conducting literature reviews?

RQ3: How do global educational institutions balance the opportunities, risks, and ethical considerations associated with ChatGPT and other AI tools?

RQ4: How can ChatGPT and other AI tools enhance educators' professional development in teaching preparation and assessment, and what are the best practices for constructing prompts to optimize their effectiveness in these areas?

3.2. Database and search strategy

Following the construction of the research questions, the document search phase has been initiated. This involved consulting various databases for relevant literature. The databases utilized in this study comprised: Science Direct, IEEE, Web of Science, Scopus, and Springer, were systematically searched in four stages: identification, screening, eligibility, and inclusion (Fig. 4). The research string was set as follows:

Research string (RS) = TITLE-ABS-KEY ((AI OR GPT) AND (ChatGPT for education professionals OR implementation of ChatGPT in education field) OR (ChatGPT in research OR AI for academic writing) OR (academic writing with ChatGPT) OR (traditional teaching method) OR (AI in teaching preparation and assessment OR ChatGPT prompts in education) OR (ChatGPT in the higher education sectors)).

The search for relevant articles involved the use of specific keywords related to AI, ChatGPT in education fields, AI in education, and ChatGPT

in higher education domains. Boolean operators such as 'AND,' and 'OR' were employed to refine the search results and generate both broad and narrow findings.

3.3. Inclusion and exclusion criteria

The data for the literature review and analysis covers the period from December 2022 to October 2024 because ChatGPT was launched by OpenAI on November 30, 2022. A two-year duration is typically adequate for executing a systematic review. The duration needed may fluctuate depending on factors such as the difficulty of the research issue, the quantity of literature, and the resources at hand. Certain systematic reviews may be finalized in as few as six months, whilst others could require 18 months or longer [37]. Table 3 provides a summary of the inclusion and exclusion criteria for selecting articles.

3.4. Data extraction and analysis

Throughout this systematic review, the authors have diligently followed the guidelines and principles set forth by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [38]. The authors aimed to minimize bias, enhance the reproducibility of the study, and maximize the validity and reliability of the findings. Moreover, rigorous methods for data extraction and synthesis have been employed, obeying predefined criteria and protocols to minimize subjectivity and ensure consistency. In order to mitigate biases during the study selection process, the review process was conducted independently by two authors who screened each study for inclusion or exclusion, with any discrepancies resolved through discussion in addition to consultation with a third reviewer. To enhance transparency in our analysis, each stage of the study selection process has been documented using the PRISMA flow diagram, which visually depicts the number of studies included or excluded at each step. Fig. 4 offers a PRISMA flowchart of the article selection process, while Table 10 in Appendix A displays the complete list of articles that were chosen for this search after the criteria were applied and each article was categorized according to the reference, methodology, publication year, key findings, and category.

4. Findings

This section presents the key findings derived from the systematic literature review, addressing the four research questions outlined in the study. Each research question is explored in detail.

4.1. RQ1: what is the comparative effectiveness of AI-supported education methods versus traditional teaching methods in enhancing student academic performance considering their respective advantages and disadvantages?

The first research question of this study aims to systematically evaluate both the advantages and disadvantages of AI-supported methods versus traditional ones. Existing studies such as [8,39] tend to emphasize the benefits of ChatGPT and AI integration but may not thoroughly explore the potential disadvantages, such as over-reliance on AI, issues with critical thinking, or limitations in AI's ability to foster deeper student engagement. Although some recent studies [40–42] offer comparisons, they focus primarily on ChatGPT effectiveness and AI-supported in specific educational contexts. In contrast, this research question encompasses not only ChatGPT but the wider spectrum of AI-supported educational tools, allowing for a broader and more holistic comparison across various AI technologies by comparing traditional methods with AI-supported approaches in areas like data analysis, teaching materials, language learning, and plagiarism prevention, further outlines the key advantages and disadvantages of both AI-supported education and traditional teaching methods, providing a

Table 2
Research questions along with their respective purposes and aims.

Research Question	Purpose of the Question	Aims
RQ1:	To assess the impact of AI-supported education on student outcomes compared to traditional approaches.	To identify the specific benefits and limitations of AI-supported education versus traditional teaching methods in educational contexts.
RQ2:	To evaluate the utility of ChatGPT in academic research and writing tasks.	Explore the potential of AI, particularly ChatGPT, to enhance the quality and efficiency of academic work mainly writing and literature reviews.
RQ3:	Explore the policies and strategies higher education institutions use to manage the integration of ChatGPT and other AI tools.	Investigate how educational institutions can leverage ChatGPT and other AI tools to enhance learning while mitigating risks to academic integrity, data privacy, and education quality, and inform the creation of balanced policies that promote innovation while safeguarding educational standards.
RQ4:	Investigate how ChatGPT and other AI technologies can be used to enhance the efficiency and effectiveness of teaching preparation and assessment practices among educators.	Identify ways in which ChatGPT can support educators in creating lesson plans, assessments, and feedback mechanisms, thereby improving teaching quality and student learning outcomes.
	Identify the best practices for constructing prompts in ChatGPT to optimize its use in teaching preparation and assessment.	Ensure that educators can maximize the benefits of ChatGPT by using well-crafted prompts that align with their specific instructional goals and educational strategies.

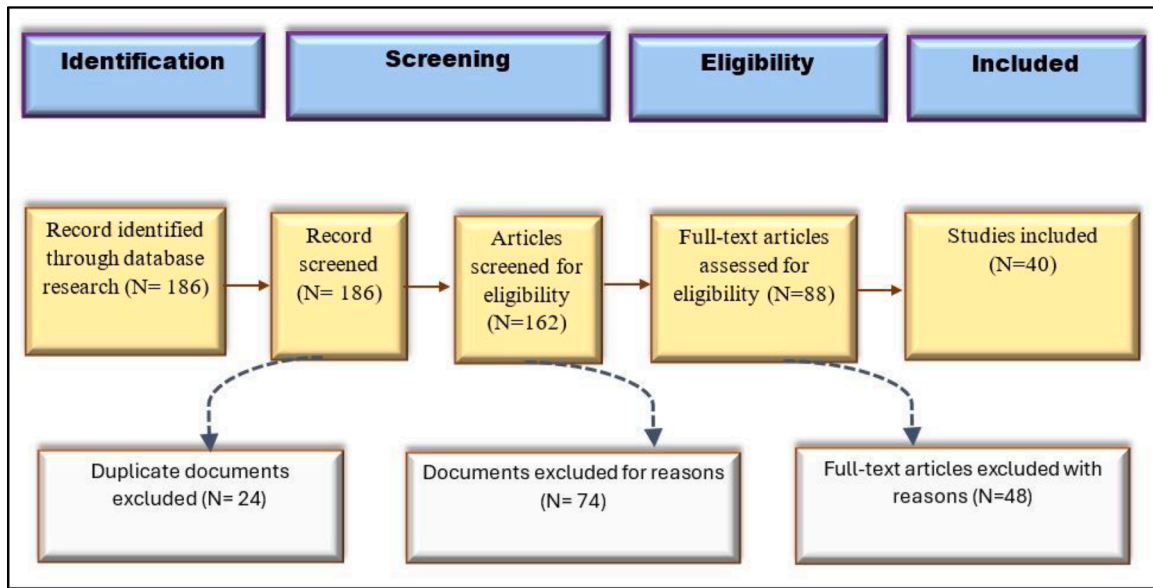


Fig. 4. PRISMA flow chart of the study selection process.

Table 3
Inclusion and Exclusion Criteria.

Criterion	Inclusion Criteria (IC)	Exclusion Criteria (EC)
Language	English must be the language used in the paper.	Not published in the English language.
Article type	Journal articles, review articles, & conference papers	Blogs, unauthentic essays, preprints, and books.
Article title	Articles with the terms "ChatGPT in education" or "AI in education" in their title or abstract.	Articles lacking the terms "ChatGPT in education" or "AI in education" in their title or abstract.
Article topic	Articles specifically discussing ChatGPT and other AI technologies integration in the education domain.	Articles discussing broader topics like GPT, LLM or quantum computing, excluding ChatGPT and AI.
Time frame	Articles published until October 2024.	< December 2022

balanced and critical perspective.

Moreover, selecting these specific applications (Fig. 5) provides a comprehensive view of how AI is influencing key areas of education. Furthermore, each of these applications significantly enhances educational processes and resolves critical issues for educators and students [43]. The subsections below discuss the four categories across AI-powered applications in the field of education.

1. Data analysis: Effective data analysis is critical for comprehending educational outcomes and enhancing instructional

approaches [29]. AI-powered systems such as RapidMiner, Qualtrics, and IBM Watson allow educators and researchers to evaluate massive amounts of data more efficiently than traditional techniques, resulting in deeper insights and more accurate predictions [44]. IBM Watson is an advanced AI system that uses techniques like NLP and ML to answer questions [45]. RapidMiner offers a user-friendly platform that allows users to build custom surveys, distribute them online or offline, and analyze the collected data through reports and visualizations [46]. This aids in spotting patterns, optimizing teaching tactics, and tailoring learning experiences. But the magic does not end there. AI can also anticipate future performance using prior data, thus serving as a fortune teller [47]. By assessing a student's trajectory, teachers may anticipate probable obstacles and modify their lessons to avoid these impediments. It's akin to possessing a predictive tool that not only identifies the current needs of students but also predicts their needs for the coming week, month, or even year. AI, like NLP techniques, can offer relevant and up-to-date materials by looking at educational trends and student preferences. By comparing student performance to projected learning objectives and identifying gaps in the curriculum, AI can also assist in creating a curriculum that meets current needs. This lets the curriculum be changed.

2. Teaching Materials and Assessments: The creation and evaluation of teaching materials are crucial to effective education. The literature contains a multitude of AI systems that offer customized learning experiences and automated grading., such as: (1) Khan Academy significantly enhances the efficiency of educational delivery and

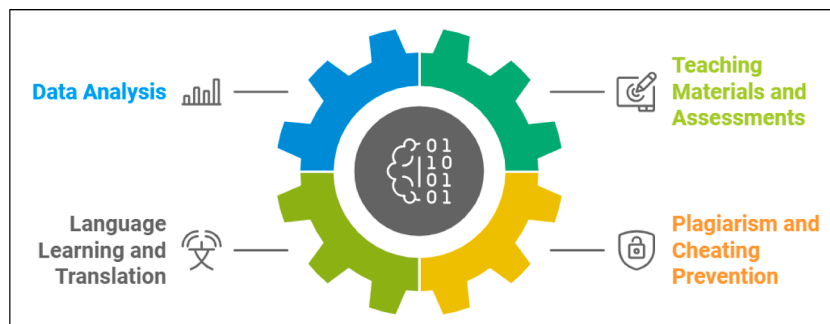


Fig. 5. Primary categories across AI-powered applications in the field of education.

assessment [48]. (2) Quillionz, which employs AI to generate assessments, including quizzes and multiple-choice inquiries, as well as summaries derived from educational content [49]. (3) Kahoot!: An interactive platform that leverages AI to assist educators in making attractive quizzes and games [50]. It examines student responses to yield insights into their comprehension levels. (4) Nearpod: An AI-enhanced instrument that enables educators to design interactive lessons featuring real-time evaluations, polls, and quizzes [51]. (5) Smart Sparrow: A personalized e-learning platform that utilizes AI to deliver tailored feedback and modifies content difficulty in accordance with student performance metrics [52]. (6) Gradescope: This AI-enabled tool aids educators in the efficient evaluation of assignments and examinations by automating the grading process and furnishing comprehensive analytics regarding student performance [53]. (7) Edulastic: An AI-empowered assessment platform that permits educators to develop and execute formative evaluations [54]. (8) Socrative: An AI-centric tool that assists educators in designing immediate quizzes and assessments, delivering prompt feedback and performance analysis [55], and finally (9) ChatGPT, that can generate questions, moving beyond the multiple-choice format to more interactive and nuanced forms of assessment [56]. These technologies and platforms enable adaptive learning, ensuring the tailoring of educational information to each student's individual needs and learning pace.

3. Language Learning and Translation: In recent times, the language learning landscape has witnessed the emergence of innovative platforms called "Speak" which harness the technological prowess of OpenAI. The public can access this newly introduced application at usespeak.com, which marks a significant advancement in language learning methodology by emphasizing aspects of language learning that traditional education settings often overlook [57]. Unlike traditional approaches, Speak offers users an immersive oral conversational experience that closely mimics interactions with

native speakers. Duolingo is a popular language learning app that uses AI to personalize lessons for each user [58], while Google Translate is a widely-used translation tool leveraging AI and ML to provide translations between multiple languages, supporting text, speech, image, and real-time video translations. Similarly, Rosetta Stone employs AI-driven speech recognition technology to help users learn new languages through immersive experiences that focus on speaking, listening, reading, and writing [59]. Babbel is another language learning platform that utilizes AI to create personalized learning paths [60], and lastly, Lingvist is an AI-driven app specifically designed to enhance vocabulary acquisition [61].

In comparison with ChatGPT, research conducted by [62] to assess AI language learning and translations, particularly ChatGPT, revealed that three out of five academic translators believed a text translated with ChatGPT was done by a human and considered it of high quality. In the language translation domain, we instruct ChatGPT to translate the English expression "Hello, how are you today?" into four distinct languages: French, Spanish, German, and Mandarin Chinese. The ChatGPT perfectly generates responses with the translated phrases in each requested language. Users can confidently communicate in multiple languages thanks to the generation of translations that are coherent and reflect the true essence of the original phrase. On the other hand, we prompt ChatGPT, "Please suggest three ways to translate the following into English" ChatGPT produces three English translations that differ in wording and structure yet effectively convey the original meaning, and the outputs maintain a similar level of clarity and naturalness as shown in Fig. 6 below.

近年來，隨著網路威脅的興起，我們確保所有資源都是乾淨的。我們掃描上傳到我們平台的每個文件，並確保其中沒有嵌入惡意軟體。為什麼？您的安全是我們的第一要務！不相信我們？親自檢查一下。如果您發現任何損壞的鏈接，請隨時聯繫我們的工作人員。我們將立即採取行動。

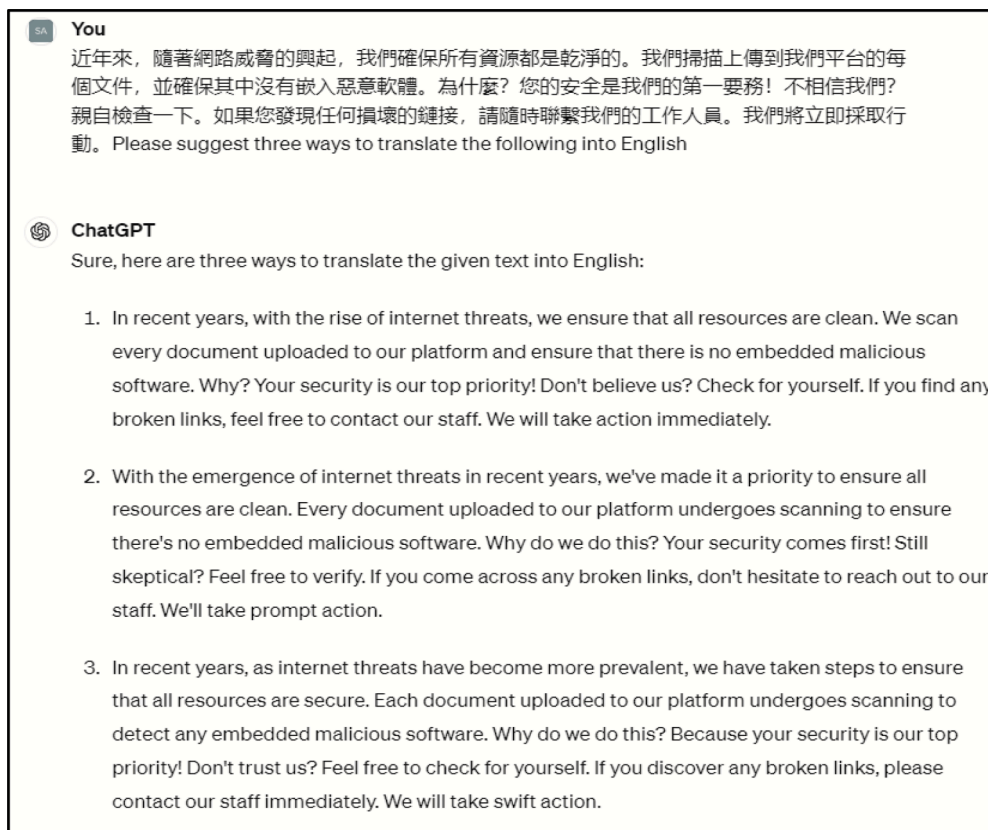


Fig. 6. Screenshot of translating traditional Chinese language to English generated by ChatGPT.

4. Plagiarism and Cheating Prevention: Both AI and traditional methods have their strengths and weaknesses in preventing plagiarism and cheating; however, the most effective approach may involve a combination of both [63]. One of the most well-known AI-powered plagiarism detection tools is Turnitin, which scans student submissions against a database of academic papers, online content, and previously submitted assignments to identify matching text [64]. Another AI tool, Grammarly, not only checks for grammatical errors but also offers plagiarism detection features [65]. Unicheck is an academic integrity tool that helps educators detect plagiarism by comparing student submissions against a vast database of sources [66]. Similarly, Copyscape is a widely used plagiarism detection tool that primarily identifies duplicate content on the internet [67]. Proctorio, on the other hand, is an AI-driven online proctoring service designed to prevent cheating during online exams by monitoring students through webcams and microphones flagging any suspicious behavior [68]. Additionally, SafeAssign is a plagiarism detection tool integrated into the Blackboard Learning Management System (LMS), which checks student submissions against a comprehensive database of academic papers. Comparably [69], Understanding the distinctions between traditional methods and their AI-powered counterparts might provide useful insights into the impact of these breakthroughs.

Table 4 below provides a detailed comparison between traditional methods and AI-supported approaches in primary education applications regarding their impacts on educational practices and student performance. While the emphasis of traditional methods relies on human contact, flexibility, and critical thinking, the approaches supported by AI

are based on automation, personalization, and data-driven insights. Each of these approaches is founded on some assumptions and accompanied by some limitations; these are discussed below.

1. Assumptions

a. AI-Supported Approaches

- Depend upon technology availability, like a stable internet connection and other digital resources
- Requires training for both educators and students in how best to make use of these AI tools.
- Presuppose safe systems for sensitive data.

b. Traditional Approaches

- Presuppose educators will have the competence and time for personalized attention
- Use methods that are standardized without making consideration for varied learning needs.

2. Limitations

a. AI -Supported Approaches:

- Excessive dependence on technology can result in underdevelopment of critical thinking and problem-solving skills.
- Data privacy and security issues demand sound policy frameworks.
- The digital divide may lead to exclusion of students from economically weaker sections.

b. Traditional Approaches

- Scalability is limited and the process for personalized learning is time-consuming.
- Resistance to changing educational needs that are happening rapidly.

Table 4

Summary of traditional versus AI-supported approaches across primary education applications, and their respective impacts on educational practices and students' performance.

Application	Traditional Methods	AI-Supported Approaches	Impact on Educational Practices	Impact on Students' Performance
Data Analysis	• Researchers manually collect data from surveys and experiments, analyze it using standard statistical methods, and form hypotheses based on the data trends. • A researcher manually reviews hundreds of academic papers to identify emerging trends and gaps in the literature.	• Qualtrics: Automatically distributes surveys to targeted participants, applies ML models for predictive analytics, and identifies patterns and correlations. • RapidMiner: Analyzes demographic and social data. • IBM Watson: Analyze thousands of research papers, extracting and categorizing data to identify trends, correlations, and emerging topics in the field.	• Increased efficiency in data collection and analysis. • Enhanced ability to identify patterns and trends.	• More accurate and timely feedback. • Greater access to diverse data sources and insights.
Teaching Materials and Assessments	• Create lesson plans and exams by hand. • Educators manually create questions of varying types and difficulties. • Uniform assessments for all students. • Fixed difficulty level for all students. • Manual grading of assignments and tests.	• Khan Academy: personalized learning materials and adaptive assessments that automate grading tools. • Gemnai: Generates custom questions. An educator can generate a variety of questions (multiple-choice, short-answer, and essay questions). • Quizlet: (LMS) Platforms use AI to create and customize assessments. • DreamBox: adaptive learning technologies to adjust the difficulty of questions in real-time based on the student's performance. • Gradescope: Automated grading.	• Personalized and adaptive learning experiences. • Streamlined grading and assessment processes.	• Customized learning experiences based on individual needs. • Faster feedback and assessment results.
Language Learning and Translation	• Language learning involves textbooks, classroom instruction, and practice with native speakers. • Translation requires human translators.	• Duolingo: real-time feedback, including speaking; listening, reading, and writing tasks that are tailored to the user's current level. • Google AI translation tools: provide instant translation services.	• More interactive and tailored language learning experiences. • Faster and more accessible translation services.	• More engaging and efficient language learning. • Easier communication and understanding of foreign languages.
Plagiarism and Cheating Prevention	• Educators manually compare student work against known sources and use intuition to detect potential cheating. • Human reviewers for assessment of authenticity • Distinguishing AI-generated content from humans.	• Turnitin: plagiarism detection scans academic content to detect similarities. • AI proctoring systems: monitor online exams by using AI-based facial recognition and behavioural analysis.	• Improved detection of plagiarism and cheating. • Enhanced exam security through AI monitoring.	• Increased academic integrity. • Reduced likelihood of cheating or plagiarism.

Table 5 presents the performance of both approaches regarding key educational applications. The clear advantages of AI-supported methods in terms of efficiency, feedback, and scalability contrast with the advantages of traditional methods related to fostering emotional connections, holistic development, and adaptability.

Table 5
AI-supported Education Vs Traditional teaching: advantages and disadvantages.

AI-Supported Education Advantages	AI-Supported Education Disadvantages	Traditional Teaching Advantages	Traditional Teaching Disadvantages
Personalized Learning: Adapt to individual student needs, providing customized learning paths, pacing, and content difficulty.	Data Privacy & Security: Potential issues with data privacy and security due to the collection of student information.	Human Connection & Social-Emotional Learning: Traditional classrooms provide important social interaction and build relationships between teachers and students, fostering a supportive learning environment crucial for social-emotional development.	One-size-fits-all Approach: This may not cater to the individual needs and learning styles of every student.
Immediate Feedback: AI tools can provide instant feedback on assignments, quizzes, and writing, allowing students to correct mistakes and learn more effectively.	Digital Divide: Risk of widening the digital divide, as not all students have equal access to technology or the internet.	Adaptive Teaching & Complex Problem Solving: Experienced teachers can adapt their teaching styles in real time based on subtle cues from students. They are also better at guiding students through nuanced problem-solving and critical-thinking tasks.	Time-Consuming: Can be time-consuming for teachers to manage large classes and provide personalized attention.
Accessibility & Flexibility: AI-powered platforms offer 24/7 access to learning materials, making education more flexible and accessible to diverse learners.	Over-reliance on AI: Could reduce opportunities for developing critical thinking and interpersonal skills.	Holistic Development: Traditional education often includes activities beyond core subjects, such as arts, sports, and extracurriculars, contributing to well-rounded development.	Outdated Methods: Potential for outdated teaching methods that may not engage modern students as effectively as digital tools.
Data-Driven Insights: AI can analyze student performance data to identify areas where they are struggling and suggest tailored interventions.	Lack of Human Touch: AI lacks the emotional connection that can be crucial in education.		
Reduced Teacher Workload: AI can automate tasks like grading, freeing up teachers to focus on individualized instruction and student interaction.			

Based on the preceding comparison, the author concludes that a well-designed blended learning approach that leverages the strengths of both human educators and AI technologies is likely to be the most effective way to improve student outcomes in the future. Notably, AI has the potential to revolutionize education, but it's not a magic bullet, rather than replacing teachers, AI solutions like ChatGPT work best when used to supplement and assist traditional teaching techniques [8,39]. Indeed, there is a risk of over-reliance on technology, which may impair the development of crucial abilities such as critical thinking, problem-solving, and interpersonal communication [70].

4.2. Proposed blended learning model

Our findings on this question strongly suggest a blended learning model combining the strengths of both approaches. AI-supported tools can extend traditional teaching to automate routine tasks, enable personalized learning, and provide insights based on data. Yet, there is a limitation to completely disregarding human judgment and emotional touch. To further justify the applicability of this method, we have examined its use in real-world scenarios such as adaptive learning platforms for K-12 education and AI-driven student assessment systems in higher education. These applications highlight how AI tools, such as personalized learning dashboards, can improve student engagement and academic outcomes. For instance, AI systems can recommend tailored learning paths based on performance data while educators concentrate on facilitating critical thinking and interpersonal skills through interactive sessions. The hybrid approach balances out the weakness of each method effectively in providing a holistic learning experience among students. Further, it recognizes limitations, including potential points of weakness, such as integration challenges of AI tools and systems with main educational ones, and the possible widening of the digital gap. These issues, however, can be addressed by having structured implementation plans, such as piloting in various educational settings and providing professional training for teachers to enhance competency in AI technologies. The implementation of guidelines and training on the use of AI will reduce risks associated with overdependence and privacy while enhancing the potential benefits of the technologies.

This blended learning model bridges the gap between AI-supported learning and traditional teaching methods, hence making it highly scalable and adaptable to changing modern educational needs.

The following Table 5 compares AI-supported education tools to traditional teaching tools, as well as their associated advantages and disadvantages.

4.3. RQ2: how does ChatGPT demonstrate its capabilities and effectiveness in academic writing and conducting literature reviews?

In contrast to previous studies, including [11,71,72] which extended the conversation on ChatGPT as a writing assistant in higher education, they did not fully address how ChatGPT demonstrates its writing capabilities and effectiveness, especially in generating coherent literature reviews. The proposed RQ2 will fill this gap by analyzing ChatGPT's potential to streamline academic writing processes more systematically. Similar to the previous studies, the work of [73] was limited to the earlier version of ChatGPT and did not provide a thorough examination of its specific capabilities in conducting literature reviews or how it enhances research efficiency. With the release of GPT-4, this review would give readers a look ahead at what ChatGPT can do in these important areas of scientific research (Fig. 7 and Table 6). It would do this by looking into how well ChatGPT does at writing literature reviews, which is an important part of academic writing that is often overlooked.

4.4. A- Academic writing

It is widely acknowledged that writing can pose challenges for many

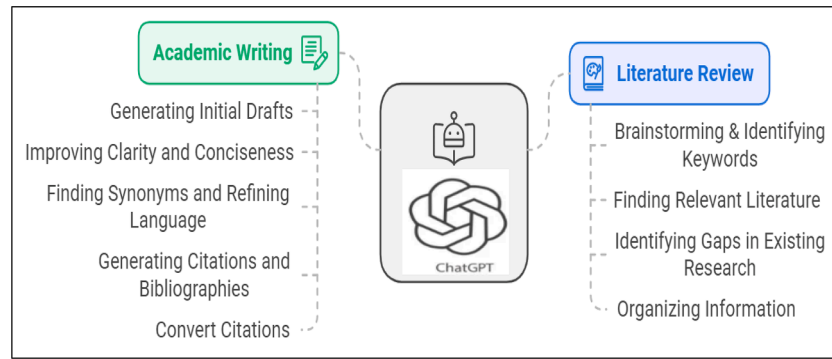


Fig. 7. Capabilities of ChatGPT in academic writing and literature reviews.

Table 6

A summary of ChatGPT's capabilities in literature reviews and academic writing.

Domain	Task	Capabilities description
Literature Review	Brainstorming & Identifying Keywords:	Input a broad research question or topic into ChatGPT and request that it create related subtopics, relevant keywords, or possible research questions. This can help to refine research emphasis and reveal avenues that may have been overlooked.
	Finding Relevant Literature:	Inputting a question for papers, books, or authors to a specific research field into ChatGPT. It will assist in identifying introductory resources.
	Identifying Gaps in Existing Research:	Once the literature, input a question into ChatGPT to examine the existing research for any gaps, disagreements, or areas that need further exploration. This can help find a study niche.
	Organizing Information:	Assist in organizing the literature obtained by creating basic outlines or mind maps. Input essential topics and concepts and ask them to generate a hierarchical structure.
Academic Writing	Generating Initial Drafts and Overcoming Writer's Block:	Provide ChatGPT with an overview or bullet points and request that it develop a draft paragraph or section. This can help overcome writer's block and start your ideas flowing.
	Improving Clarity and Conciseness:	Paste a paragraph or sentence into ChatGPT and request that it be rephrased for clarity, conciseness, or a more scholarly tone. This can be very beneficial for non-native English speakers.
	Finding Synonyms and Refining Language:	Ask ChatGPT for synonyms or alternative phrases to reduce repetition and add variation.
	Generating Citations and Bibliographies (with Caution):	Write the prompt into ChatGPT to create citations for a given topic. It's critical to double-check their accuracy against a respected citation style guide and database.
	Convert citations	Paste the citation in any format and ask ChatGPT to convert it to the required format.

people [74,75]. AI-powered tools like ChatGPT offer a quick solution by generating diverse versions of text, which can be a valuable source of inspiration for authors grappling with writer's block [71]. Since its introduction, ChatGPT has been widely employed for a range of writing tasks, including creating essays, drafting applications, generating email content, and composing research papers, as evidenced by its extensive

usage [76,77]. Compared to earlier works, the author of [78] leveraged ChatGPT to compose an article related to medical topics, arguing that the future of medical writing will heavily rely on AI-assisted tools. Similarly, Tai [79] emphasized the importance of using ChatGPT for accurate medical writing and addressed associated concerns. Additionally, [80,81] highlighted the substantial assistance AI-powered tools can offer in producing medical content and their potential role in shaping its path. However, it is essential to recognize that human oversight remains crucial for validating ChatGPT's outputs. Consistent with this perspective, Kumar [82] highlighted the significant potential of ChatGPT in academic writing, particularly when complemented by human guidance. These insights collectively emphasize the promising prospects of integrating ChatGPT into writing processes while underscoring the indispensable role of human judgment in ensuring precision and credibility.

In the context of academic writing, we attempted to utilize ChatGPT for a prompt (see Fig. 8 "Examine the role of citizen science projects in advancing scientific research and education, discussing examples of successful initiatives across various disciplines and their impact on public engagement with science." Surprisingly, this prompt led ChatGPT to elaborately discuss how citizen science projects involve the public in scientific research, contributing to data collection, analysis, and dissemination. It can explore examples such as crowd-sourced biodiversity monitoring, astronomical observations, and environmental monitoring initiatives. Additionally, ChatGPT can discuss the benefits of citizen science in fostering public understanding of science and promoting scientific literacy.

In the realm of reference formatting and conversion, ChatGPT, for instance, can explain the subtle differences between in-text citations for a paraphrase and a direct quote in APA or the correct way to format a website citation in MLA. We extend a prompt to ChatGPT to convert the reference from the IEEE style to the Chicago style (Fig. 9).

4.5. B- Literature review

ChatGPT's usefulness extends to enhance language and improve the ability to articulate and communicate research ideas and findings, thereby advancing the publication process and facilitating quicker availability of research results [80,83,84]. ChatGPT has proved its utility in assisting the creation of inquiries for comprehensive systematic reviews with high precision, as evidenced by Wang et al. [85]. During literature review and synthesis, ChatGPT excels at extracting and analyzing information through text extraction and semantic analysis, and synthesizing literature by creating summary tables, performing comparative analysis, and automating literature synthesis. Imran et al. [11] found that researchers can derive significant advantages from using ChatGPT because it can assist in identifying relevant literature. Researchers can achieve this by generating comprehensive summaries of articles or providing a comprehensive list of papers related to a specific topic or keyword. This invaluable feature not only simplifies the task of

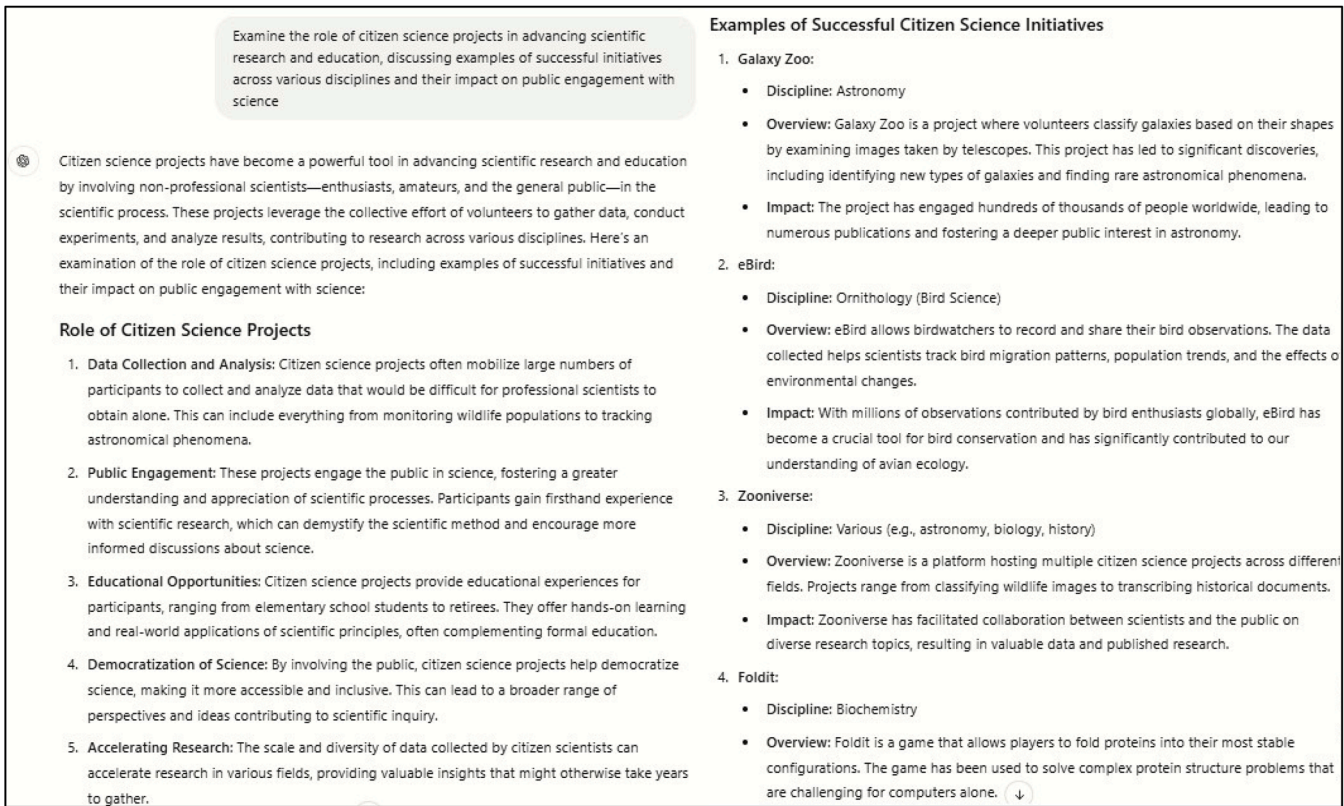


Fig. 8. Screenshot of citizen science projects generated by ChatGPT.

going through a large volume of literature but also ensures that researchers have the necessary resources to improve the quality and depth of their research efforts. Additionally, the author of [72,86] discussed how ChatGPT can assist researchers in conducting literature reviews by summarizing existing research, identifying key findings, and potentially even generating insights or recommendations based on the synthesized information, and he concludes that ChatGPT may not hold up under the analysis of a comprehensive literature review due to its tendency for generating an excessive number of counterfeit academic papers, thus rendering it exceedingly challenging to even consider recommending ChatGPT as a collaborative writing assistant. The research findings of [71,87] indicate that ChatGPT as a tool for literature review assistance aids researchers in extracting relevant information related to their research topic with an overall accuracy rate of 70 %. Furthermore, it is demonstrated that ChatGPT offers a more economical and efficient approach to achieving this level of accuracy compared to human researchers. Gao et al. [88] conducted an experiment using 50 scientific journal abstracts to create AI-generated abstracts, namely ChatGPT. The results accurately identified 68 % of machine-generated abstracts but incorrectly attributed 14 % to ChatGPT. The results showed that human reviewers found it difficult to distinguish between abstracts generated by ChatGPT and those authored by humans, indicating its proficiency in handling complex textual outputs. However, it is concluded that while ChatGPT shows promise in providing a quick and cost-effective overview of a subject, it cannot currently generate original research ideas or critically evaluate existing literature to the same extent a human researcher can [74]. In academic settings, particularly literature reviews, ChatGPT is frequently utilized to summarize previous research, provide bibliographies, and offer insights into upcoming trends. Nonetheless, a notable constraint has arisen due to the AI's propensity to generate "hallucinations" the fabrication of non-existent papers, references, or data that lack correspondence with actual sources, which may present significant threats to academic integrity and precision [89]. ChatGPT often generates incorrect or non-existent references. Example:

Fig. 10 references were checked, and the first reference does not exist. Furthermore, Gammoh [90] advocates for human control in the use of ChatGPT for academic writing. They assert that unverified fake references may mislead readers and undermine the overall integrity of scientific work. They argue that these hallucinations highlight the limitations of AI's data sources and processing models, trained on extensive yet finite datasets that might not include the most recent or all academic publications.

In the literature review, the prompt to ChatGPT: "Conduct a literature review with an in-text citation on recent advancements in the field of nanotechnology, focusing on applications in medicine, electronics, and environmental science." has been utilized (Fig. 10). Remarkably, this prompt guides ChatGPT to summarize recent research articles, reviews, and studies related to nanotechnology. It also covers topics such as nanoparticle-based drug delivery systems, Nano-electronics, Nano-sensors for environmental monitoring, and other emerging applications. Moreover, the literature review discusses challenges, future directions, and potential ethical considerations in the field of nanotechnology, along with in-text citations.

4.6. RQ3: how do global educational institutions balance the opportunities, risks, and ethical considerations associated with ChatGPT and other AI tools?

The third RQ of this study presents a broader perspective that addresses several gaps left unexplored by the studies of [9,43,91] that focused on specific elements of ChatGPT's impact on higher education, such as academic integrity or mitigation strategies, but did not systematically examine how institutions globally are navigating the nuanced balance between the benefits and challenges of AI tools. Furthermore, earlier studies such as [92] have been limited to how ChatGPT is disrupting higher education, concentrating specifically on Australia, New Zealand, the United States, and the United Kingdom. Additionally, the previous studies [91,93,94] focused on specific aspects

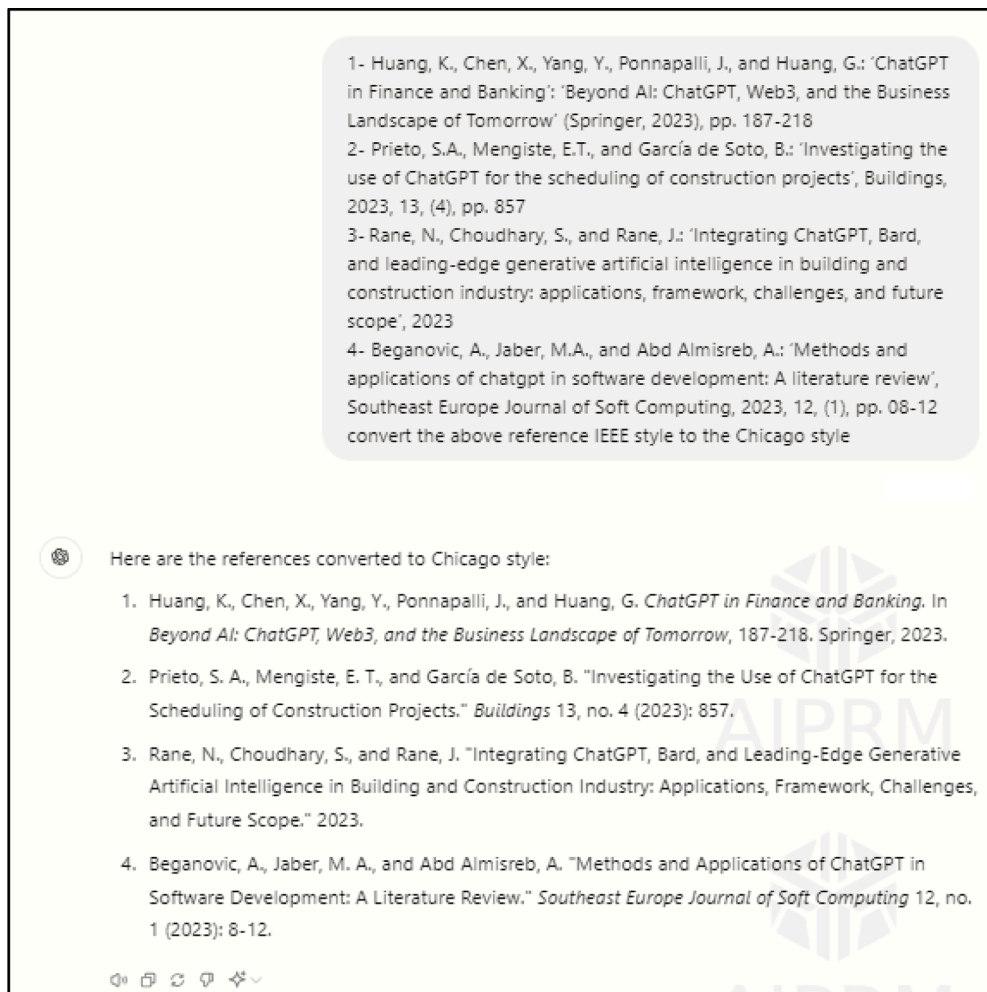


Fig. 9. Screenshot generated by ChatGPT for references converted from IEEE to Chicago style.

like mitigation strategies to maintain academic integrity. The proposed RQ3 hereby addresses the multifaceted and global nature of balancing AI in education, filling gaps in the existing literature by providing a comprehensive, institution-level analysis that explores a global perspective, measures, policy development stage, integration level, and focus areas that will offer unique insights into how educational institutions globally are addressing the dual challenges of innovation and integrity in the AI era. The subsection below highlights the measures and implications that have been employed by educational institutions around the globe to adapt to AI's evolving role, while Fig. 11 below depicts global educational institutions' names.

To understand the impact of integrating ChatGPT and other AI technologies into education, one must carefully consider how it affects the maintenance of academic integrity and the effective detection of plagiarism. Before deliberating on the potential academic prohibition of AI, it is imperative to precisely assess the challenges presently confronting education and the profound influence of artificial intelligence therein. According to [14], from a global standpoint, the education sector grapples with numerous issues, including the inequitable allocation of educational resources, inconsistent quality of education, inadequate educational content, antiquated pedagogical methods, excessive student workloads, and a flawed educational evaluation system [95]. These issues significantly impede the equity and quality of education, obstructing its advancement and evolution. On the other hand, the only issue with the use of ChatGPT in higher education is the risk of plagiarism [43,96]. Understanding this aspect is crucial for educators, policymakers, and stakeholders to make informed decisions about the use of

AI-driven tools in education [17,43].

Global educational institutions have responded to ChatGPT and similar AI language models with a mixture of caution, curiosity, and proactive adaptation with the adoption of various strategies to balance the challenges and opportunities presented by AI technologies like ChatGPT while maintaining academic standards [96,97]. One such institution, the Massachusetts Institute of Technology (MIT), has embraced the potential of AI by incorporating it into its curriculum [98]. MIT's Media Lab, for instance, is actively exploring the role of AI in education, emphasizing the importance of teaching students to work alongside AI as collaborators rather than mere recipients of information [1]. Seattle public schools prohibited the utilization of Chat GPT in January 2023, while Sciences Po in Paris underscored the ban on academics employing Chat GPT or other AI technologies in secret ways [99]. RV University in Bangalore, India, has instituted a stringent prohibition against students using Chat GPT for work completion, examination participation, or laboratory assessments [100]. Elsen-Rooney [101] reported that the New York City Education Department (NYC) has prohibited access to ChatGPT on school devices and networks, therefore restricting students and instructors from utilizing the platform. New South Wales was the inaugural area in Australia to prohibit students from using Chat GPT, followed by Queensland, Tasmania, and Western Australia, which have implemented such restrictions in public schools to uphold scholastic standards [93,102]. The University of California, Berkeley, has developed guidelines to discourage students from using such tools to complete their assignments, highlighting the importance of original thought and intellectual honesty in academic pursuits. These

Literature Review: Recent Advancements in Nanotechnology and Its Applications 1. Introduction Nanotechnology, the manipulation of matter at the nanoscale (1 to 100 nanometers), has emerged as a transformative field with vast applications across various domains. This literature review examines the recent advancements in nanotechnology, particularly focusing on its applications in medicine, electronics, and environmental science. 2. Nanotechnology in Medicine Nanotechnology has revolutionized the medical field by enabling new diagnostic and therapeutic approaches. 2.1 Drug Delivery Systems One of the most significant advancements in nanomedicine is the development of targeted drug delivery systems. Nanoparticles, such as liposomes and dendrimers, can deliver drugs directly to diseased cells, minimizing side effects and enhancing treatment efficacy. Recent studies have shown that nanoparticles can be engineered to release drugs in response to specific stimuli, such as pH or temperature, providing controlled and sustained drug delivery (Zhang et al., 2023). 2.2 Cancer Therapy Nanotechnology has also made strides in cancer therapy, with the development of nanocarriers that can deliver chemotherapy drugs directly to tumors. These nanocarriers, often coated with targeting ligands, can enhance the precision of drug delivery, reducing harm to healthy tissues. Additionally, nanomaterials like gold nanoparticles have been explored for photothermal therapy, where they are used to convert light into heat, selectively destroying cancer cells (Huang et al., 2022). 2.3 Diagnostic Imaging Nanoparticles are increasingly used in diagnostic imaging techniques, such as magnetic resonance imaging (MRI) and computed tomography (CT). For instance, superparamagnetic iron oxide nanoparticles (SPIONs) have been employed as contrast agents in MRI, improving the visualization of tumors and other abnormalities. Recent research has focused on developing multifunctional nanoparticles that combine imaging and therapeutic capabilities, known as theranostic agents (Li et al., 2024). 3. Nanotechnology in Electronics The electronics industry has benefited immensely from nanotechnology, particularly in the development of smaller, faster, and more efficient devices.	3.1 Nanoelectronics Nanoelectronics, which involves the use of nanomaterials in electronic devices, has led to significant improvements in transistor performance and energy efficiency. The miniaturization of transistors to the nanoscale has enabled the continuation of Moore's Law, with devices becoming increasingly powerful and compact. Recent advancements include the development of carbon nanotube transistors and 2D materials like graphene, which offer superior electrical properties compared to traditional silicon-based transistors (Wang et al., 2023). 3.2 Flexible and Wearable Electronics Nanotechnology has also enabled the creation of flexible and wearable electronic devices. These devices, often made from nanomaterials such as silver nanowires and graphene, are lightweight, bendable, and can be integrated into clothing or accessories. Recent innovations in this field include the development of flexible batteries and supercapacitors, which are essential for powering wearable electronics (Kim et al., 2024). 3.3 Quantum Dots Quantum dots, semiconductor nanoparticles with unique optical and electronic properties, have found applications in displays and lighting. These nanoparticles emit light of specific wavelengths when excited, making them ideal for use in high-resolution displays with vibrant colors. Recent advancements have focused on improving the stability and efficiency of quantum dots, as well as exploring their potential in quantum computing (Chen et al., 2023). 4. Nanotechnology in Environmental Science Nanotechnology offers innovative solutions to environmental challenges, particularly in pollution control and sustainability. 4.1 Water Purification Nanomaterials have shown great promise in water purification, with their ability to remove contaminants at the molecular level. Nanofiltration membranes, made from materials like graphene oxide, can efficiently filter out heavy metals, pathogens, and other pollutants from water. Recent research has explored the use of photocatalytic nanoparticles, such as titanium dioxide, which can degrade organic pollutants under light, offering a sustainable approach to water treatment (Singh et al., 2024). 4.2 Air Pollution Control Nanotechnology has been applied to air pollution control through the development of nanocatalysts that can break down harmful pollutants in the atmosphere. For example, platinum and palladium nanoparticles are used in catalytic converters to reduce emissions from vehicles. Recent advancements include the development of more efficient and cost-effective nanocatalysts, as well as the exploration of nanomaterials for	effective nanocatalysts, as well as the exploration of nanomaterials for capturing and storing carbon dioxide (CO₂) from the air (Jiang et al., 2023). 4.3 Environmental Sensing Nanotechnology has also led to the creation of highly sensitive environmental sensors. These nanosensors, often based on nanowires or nanoparticles, can detect trace amounts of pollutants, such as volatile organic compounds (VOCs) and heavy metals, in the environment. Recent studies have focused on developing portable and real-time sensing devices that can monitor environmental conditions continuously, providing valuable data for pollution control and environmental protection (Zhao et al., 2024). 5. Conclusion The field of nanotechnology continues to advance rapidly, with significant developments in medicine, electronics, and environmental science. These advancements have the potential to address some of the most pressing challenges in these fields, offering new and innovative solutions. As research progresses, the integration of nanotechnology into various industries is expected to grow, leading to further breakthroughs and applications. References - Chen, X., et al. (2023). 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Stimuli-Responsive Nanoparticles for Targeted Drug Delivery. <i>Journal of Controlled Release</i>, 350, 304-322.
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Fig. 10. A literature review on advancements in the field of nanotechnology generated by ChatGPT.

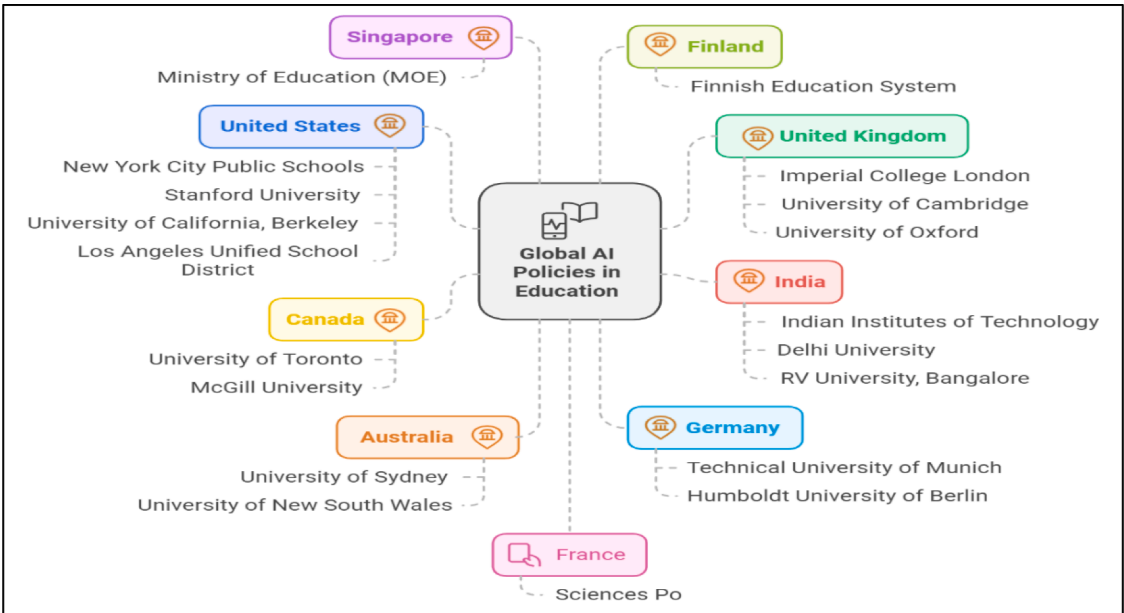


Fig. 11. Global educational institutions' policies on AI Tool Integration.

measures were taken to ensure the protection and well-being of students [97]. The execution of these steps is to guarantee that students utilize Chat GPT technologies appropriately, prevent excessive dependence and misuse, and uphold academic integrity and educational quality [94, 103].

On the other side, certain universities, such as the University of Cambridge, are adopting a balanced approach to emphasize the importance of teaching students to use technology wisely and ethically. However, organizations like the Khan Academy are taking a proactive stance by integrating ChatGPT into their platform [104]. They utilize it as an interactive learning partner to provide feedback to students, particularly when prompted effectively [105]. The University of Hong Kong mandates that students refrain from using AI technologies, like ChatGPT, without explicit written consent from instructors; failure to

comply would be considered plagiarism [14,94].

Table 7 provides the measures that have been taken by global educational institutions to regulate ChatGPT and other AI tools. It shows the policy development stage, the level of AI tool integration, and the main areas of focus. While Fig. 12 shows the approaches to AI tool usage in education such as institutional policies and practices.

From the above table, several parameters that reflect AI adoption depth and purpose determined each institution's AI tool integration level: Minimal, Moderate, or High. The University of California, Berkeley, and Finland's educational system, which uses AI to support instructors and curriculum creation, had high integration. Organizations like Imperial College London received ratings of moderate or high integration due to their organized approach to ethical AI use. New York City Public Schools, which prohibit academic dishonesty, were rated as

Table 7

Measures taken by global educational institutions to regulate ChatGPT and other AI tools.

Country	Institution	Measure	Implications	Policy Development Stage	AI Tool Integration Level	Focus Areas
United States	New York City Public Schools	Banned ChatGPT on networks and devices	Prevents students from using AI to cheat on assignments.	Established	Minimal	Academic integrity, Homework
	Stanford University	Encouraged faculty to consider AI-generated text	New guidelines being developed for AI use in classrooms	Under review	Moderate	Academic integrity, Classroom use
	University of California, Berkeley	AI tools allowed for brainstorming and drafting	Encourages critical thinking and analysis while using AI for creative tasks	Established	High	Creativity, Critical analysis
	Los Angeles Unified School District	Initially banned ChatGPT, later lifted ban	Moving towards a regulated approach, promoting responsible AI use	Evolving	Moderate	Responsible AI usage, Ethical AI
United Kingdom	Imperial College London	Permitted use with transparency and referencing	Encourages exploring AI's educational benefits while maintaining transparency	Established	High	Learning enhancement, AI literacy
	University of Cambridge	Open discussions and guidelines for responsible AI use	Promotes responsible use and deeper student engagement with AI tools	Evolving	Moderate	Responsible AI usage, Ethical thinking
	University of Oxford	AI-generated text used for student critique	Develops critical thinking skills by challenging students to critique AI outputs	Established	Moderate	Critical thinking, AI critique
Canada	University of Toronto	Banned AI in certain courses (e.g., computer science)	Addresses risks of cheating in technical courses like coding	Established	Minimal	Academic integrity, Technical skills
	McGill University	Allowed AI with emphasis on academic integrity	Allows students to engage with AI while emphasizing originality and citation	Established	Moderate	Research, Academic integrity
Australia	University of Sydney	AI tools allowed in classrooms for educational purposes	Encourages responsible AI engagement while preventing misuse for cheating	Evolving	High	Classroom learning, Responsible AI
	University of New South Wales	AI as a teaching aid	Investigates AI's role in augmenting human knowledge	Evolving	High	AI as a teaching tool, Knowledge augmentation
Singapore	Ministry of Education (MOE)	A balanced approach to AI use	Focuses on fostering creativity and critical thinking in students	Evolving	Moderate	Creativity, Skill development
India	Indian Institutes of Technology (IITs)	Mixed policies across different IITs	Some IITs focus on learning enhancement, while others emphasize integrity	Evolving	Moderate	Research, Academic integrity
	Delhi University	Warned against AI use for academic dishonesty	Reinforces the importance of originality and understanding	Established	Minimal	Academic integrity, Ethical use
	RV University, Bangalore	Banned ChatGPT	Encourages faculty to design AI-resistant assessments	Established	Minimal	Assessment, AI-resistant evaluations
Germany	Technical University of Munich	AI use allowed with strict citation guidelines	Students are required to reflect on AI limitations while using it	Established	Moderate	Critical thinking, Research, Citation
	Humboldt University of Berlin	AI integrated into courses on text analysis	Prepares students for future digital workplaces	Evolving	High	Digital literacy, Text analysis
Finland	Finnish Education System	Exploring AI to assist teachers and enhance learning	Considers AI as a tool for educational support rather than a threat	Under review	High	Teacher support, Learning enhancement
France	Sciences Po	Banned AI writing tools with severe penalties	Strictly opposes the use of AI for academic purposes, with severe consequences for violations.	Established	Minimal	Academic integrity, AI ethics
China	Tsinghua University	Using AI-generated tests to assess students	Leveraging AI to create personalized, complex exam questions	Evolving	High	Assessment, Personalized learning
United Arab Emirates	Higher Colleges of Technology	"Learn with AI" approach	Encourages ethical AI use while integrating it into the curriculum	Evolving	High	Critical thinking, Ethical AI usage

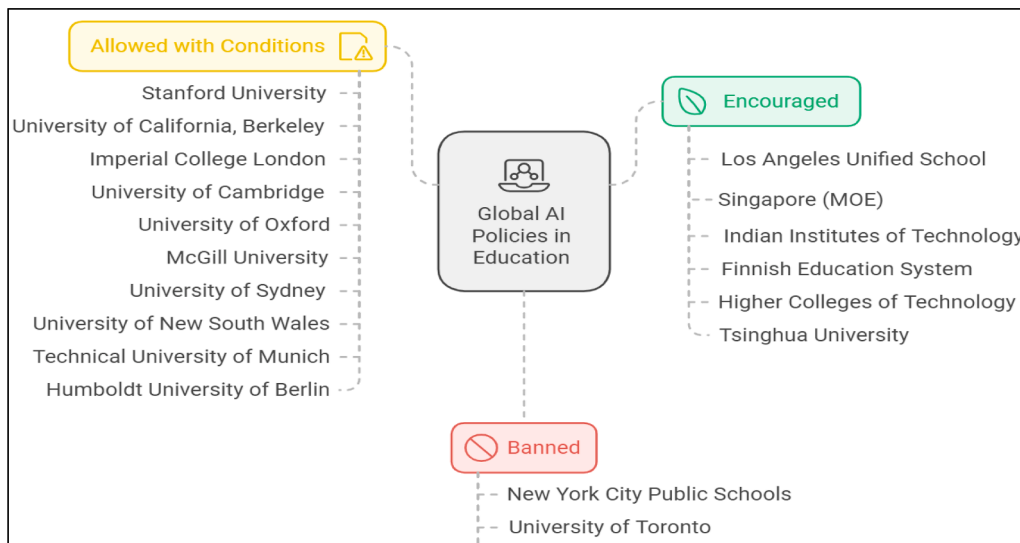


Fig. 12. Approaches to AI tool usage in education: institutional policies and practices.

minimal, indicating minimal AI use beyond restriction. This classification shows the range of AI adoption in institutions, from regulatory to educational.

In addition to the above-discussed opportunities and risks of AI tools in education, this study also analyzed the ethical considerations of AI tool usage. The findings of the analysis revealed that the ethical use of AI tools like ChatGPT in higher education has raised significant concerns, particularly regarding academic integrity, data privacy, and the potential over-reliance on AI. The increasing integration of ChatGPT and similar tools into educational environments necessitates their ethical application to prevent misuse and promote a responsible learning culture. Moreover, understanding public sentiments about Generative AI, such as perceptions of fairness, trust, and biases, is very critical in formulating policies that balance innovation with ethical responsibility. Recent studies have explored how Generative AI, such as ChatGPT, is perceived by users in tasks traditionally handled by humans, including systematic searching and sentiment analysis, revealing both enthusiasm and apprehension about its transformative impact [106,107]. The latter perceptions shape the adoption and integration of AI tools into educational and broader societal contexts. This study summarizes the ethical considerations as follows:

4.7. Academic integrity concerns

A primary ethical challenge with ChatGPT is its potential impact on academic integrity. Students may misuse the AI tool to generate essays, reports, or assignments without proper citation, leading to issues of plagiarism [108]. This weakens the traditional principles of academic honesty, which expect students to produce original work. Researchers have pointed out that without clear guidelines, students might treat AI-generated content as their own, further complicating efforts to maintain academic honesty [109]. Universities need to develop robust policies to guide the ethical use of AI tools in academia, ensuring students are educated on the responsible use of these technologies to avoid unintended misconduct [43].

4.8. Privacy and security issues

Another ethical concern is the risk to students' privacy and security. As AI tools like ChatGPT process large amounts of data, there is a potential for sensitive or personal information to be exposed or misused. The AI's ability to generate responses based on user input could lead to unauthorized use of private academic data if not properly safeguarded.

This highlights the need for severe privacy policies and practices to protect students' and faculty members' data from potential breaches [110]. Institutions should ensure that students are aware of these risks and implement necessary safeguards, including encryption and compliance with data protection regulations.

4.9. Over-Reliance on AI and cognitive development

The growing use of ChatGPT could also lead to over-reliance on AI, which may hinder students' critical thinking and problem-solving skills. If students use ChatGPT as a support, they may avoid essential cognitive processes such as reflection, creativity, and independent reasoning. AI tools should be used to complement, not replace, the educational experience, encouraging students to engage more deeply with content. Over-dependence could diminish the development of key academic and life skills, ultimately affecting students' intellectual growth and preparedness for real-world challenges [111].

Limitations in Complex Problem Solving: AI models such as ChatGPT often struggle to handle mathematical, complex, and integrated problems, especially those involving real-world scenarios across multiple systems. This limitation underscores the need for human oversight and emphasizes that AI should complement rather than replace deep problem-solving tasks [112].

4.10. Addressing plagiarism concerns within AI-generated content

A major ethical concern with ChatGPT is that it produces text indistinguishable from that produced by a human without proper acknowledgement, which borders on plagiarism. These further muddles the waters of what work is AI-generated and what is not; hence, a very valid threat to academic integrity and intellectual property. Studies have emphasized the necessity for string attribution mechanisms and user education on these risks [89]. With that in mind, institutions should establish guidelines on citing AI-generated contents and foster plagiarism detection tools specially designed for AI outputs. Moreover, raising awareness among users about the limitations of ChatGPT in recognizing and attributing human-written text would promote its responsible use and minimize accidental plagiarism.

4.11. Recommendations for ethical use of AI in education

To address the above ethical concerns, the following recommendations have been provided for the educational institutions':

1. Institutions should implement clear guidelines on the ethical use of AI tools.
2. Institutions should offer training to students and educators on responsible AI usage.
3. Universities should create mechanisms to detect and prevent cheating or plagiarism.
4. Universities should encourage transparency, ensuring students understand how AI tools like ChatGPT function and the ethical implications of their use.
5. AI tools should encourage critical thinking and human supervision rather than solely providing answers. Educational AI systems must promote active learning and ensure students engage thoughtfully with the content.

The suggested recommendations can help in maintaining academic integrity while allowing for the positive integration of AI technologies in educational settings.

4.12. RQ4: how can ChatGPT and other AI tools enhance educators' professional development in teaching preparation and assessment, and what are the best practices for constructing prompts to optimize their effectiveness in these areas?

The fourth research question of this study delves into how ChatGPT and other AI platforms can significantly enhance teacher professional development, particularly in the areas of lesson preparation and student evaluation. This investigation aims to identify the most effective ChatGPT prompts and strategies to maximize the benefits of these tools, thereby optimizing outcomes in educational settings. In comparison with previous studies such as [70,113,114], which emphasized teachers' perceptions of ChatGPT in supporting general teaching and learning processes focusing on specific educational functions like lesson planning and critical thinking. However, it doesn't directly focus on the professional development required for educators to utilize AI effectively in

teaching preparation and assessment. While the work of [93,115] looks at broader challenges and opportunities with ChatGPT vital areas for educators using AI in teaching preparation. But doesn't delve into the specific technicalities of how prompts are constructed to maximize AI's potential for professional development. This disparity underscores an important opportunity to investigate how ChatGPT can be leveraged as a tool to enhance teachers' efficiency, creativity, and effectiveness in designing instructional materials and evaluating student performance. Additionally, research on prompt construction for maximal output in these areas has been scarce.

4.13. ChatGPT in teaching preparation

Educators can streamline their preparation by leveraging ChatGPT in the lesson planning process, which can significantly enhance the efficiency and effectiveness of educators [114,115]. ChatGPT serves as an intelligent companion in the lesson-planning process [75]. Educators can leverage their vast knowledge base to brainstorm innovative ideas, gather relevant information, and construct coherent lesson outlines [116]. By simply posing a query such as "How can I teach the water cycle to a third-grade class?" or "What are some engaging activities for a high school history lesson on the Industrial Revolution?", ChatGPT generates a plethora of interactive, age-appropriate, and curriculum-aligned suggestions. The study of [114] discussed that incorporating ChatGPT into lesson planning can significantly enhance the educational experience by providing a rich repository of resources and tools for educators. The authors further explored the capabilities of ChatGPT by requesting it to prepare a 45-minute lesson plan focused on developing English interactive skills for diploma students, incorporating teamwork elements. Fig. 13 below illustrates this prompt and its response. Furthermore, ChatGPT assists in content creation by generating a variety of instructional activities that cater to different learning styles, such as visual, auditory, and kinesthetic learners [113]. Educators can specify the type of activities they need, whether it's group work, hands-on experiments, or interactive

prepare a lesson plan for 45-minutes lesson on English interactive skills for diploma student and incorporate team work pain.

Lesson Plan: Enhancing English Interactive Skills with a Focus on Teamwork

Duration:
45 minutes

Target Audience:
Diploma Students

Lesson Objectives:

1. To improve students' English interactive skills, including speaking and listening.
2. To enhance students' ability to work effectively in teams.
3. To develop confidence in using English in various interactive scenarios.

Materials Needed:

- Whiteboard and markers
- Index cards
- Timer
- Handouts with role-play scenarios
- Audio-visual equipment (if needed for playing videos)
- Sticky notes

Lesson Outline:

Introduction (5 minutes)

- **Welcome and Warm-up:** Greet students and briefly discuss the importance of interactive skills in English.
- **Icebreaker Activity:** Conduct a quick icebreaker where students introduce themselves and share one interesting fact.

Activity 1: Pair Discussion (10 minutes)

- **Instructions:** Pair up students and give each pair an index card with a discussion topic (e.g., favorite movies, recent news, travel experiences).
- **Task:** Students discuss the topic for 5 minutes, practicing active listening and speaking.
- **Debrief:** Ask a few pairs to share highlights from their discussions with the class.

Activity 2: Team Role-Play (15 minutes)

- **Instructions:** Divide students into small groups of 4-5. Provide each group with a handout detailing a scenario for role-play (e.g., a business meeting, planning a trip, organizing an event).
- **Preparation:** Groups have 5 minutes to read the scenario and assign roles.
- **Role-Play:** Each group performs their role-play for 5 minutes.
- **Feedback:** After each role-play, provide constructive feedback focusing on language use, teamwork, and interaction.

Activity 3: Problem-Solving Task (10 minutes)

- **Instructions:** Give each group a problem-solving task that requires discussion and consensus (e.g., planning a charity event, designing a marketing strategy for a product).
- **Task:** Groups discuss and come up with a solution, then present their solution to the class in 2 minutes.
- **Debrief:** Highlight effective teamwork strategies and language used during the presentations.

Conclusion (5 minutes)

- **Review:** Summarize key points learned about interactive skills and teamwork.
- **Feedback:** Use sticky notes for students to write down one thing they learned and one area they want to improve.
- **Closing:** Encourage students to practice these skills in their daily interactions and thank them for their participation.

Detailed Activities:

Icebreaker Activity

- **Objective:** To create a comfortable environment and encourage speaking.
- **Task:** Students introduce themselves and share one interesting fact.

Fig. 13. Screenshot of lesson plan generated by ChatGPT.

discussions. It can also help in creating multimodal materials such as videos, infographics, and presentations, which are essential in today's multimedia-driven educational landscape [117]. By doing so, it empowers teachers to transcend the limitations of traditional textbook-centric teaching and embrace more interactive and inclusive pedagogical strategies.

4.14. ChatGPT in assessments

The evaluation of learners has increasingly attracted attention and become a prominent subject. It is premature to conclude; nonetheless, educators may soon need to reevaluate student assessment methods. They may need to modify the present assessment methods to incorporate more novel evaluations [118]. Existing literature has shown that educators possess inadequate abilities and competencies to implement effective assessment procedures that advance learning [119]. The increasing usage of AI in professional settings necessitates the incorporation of generative AI tools in educational environments, equipping students with the skills to utilize these technologies constructively and securely, hence preparing them for success in an AI-centric workforce post-graduation. Consequently, instructors may utilize generative AI models such as ChatGPT to enhance students' learning. Certain inquiries require immediate responses, such as whether we should provide training for teachers and students on the utilization of contemporary generative AI technologies to enhance educational practices. How can we include generative AI tools in teacher education programs to equip teacher candidates or pre-service teachers for the efficient utilization of AI technologies in their classrooms? The ChatGPT model can be employed to generate quizzes with varying difficulty levels (e.g., high, medium, easy) on the same topic. For instance, as demonstrated in Fig. 14, and Fig. 15 below, ChatGPT has been tasked with creating a high-difficulty quiz on "Sorting and Searching Algorithms". Additionally, it

was requested to create quizzes of other difficulty levels, such as easy and difficult, and observed significant differences between them.

The process of developing precisely rubrics specifically designed for the assessment of project-based learning initiatives serves to significantly enhance both the clarity and consistency with which the work produced by students is evaluated and appraised [70,120]. In light of this, we prompt ChatGPT to an extensive and detailed rubric template that encompasses a variety of criteria, which include the quality of research conducted, the level of creativity demonstrated, and the effectiveness of the presentation delivered. Surprisingly a comprehensive rubric template that includes criteria for research quality, creativity, and presentation was presented as shown in Fig. 16 below.

4.15. Other AI tools in teaching preparation and assessment

In educational technology, AI-driven tools are transforming course and lesson plan creation, differentiated learning, personalized tutoring, accessibility, and digital learning [116,121]. There are a variety of tools available besides ChatGPT, each with specific features, advantages, and disadvantages. For instance, Congii is a useful technology that can continue to support assessment, provide real-time feedback, and create question papers, but it requires significant customization for certain disciplines. Although Khanmigo of Khan Academy offers tailored instruction and interactive topic exploration, it is still restricted to a small number of subjects. Coursebox facilitates the creation of lectures and the summarization of extensive knowledge, but its features for effectively interacting with other learning platforms are restricted [122]. Additionally, lesson preparations and course materials are made simpler with Top Hat and Microsoft Copilot. Although Top Hat performs a fantastic job of giving students an engaging experience through polls and quizzes, frequent internet use appears to be one of the drawbacks. Because Microsoft Copilot is generative AI-driven, it has assisted tutors in

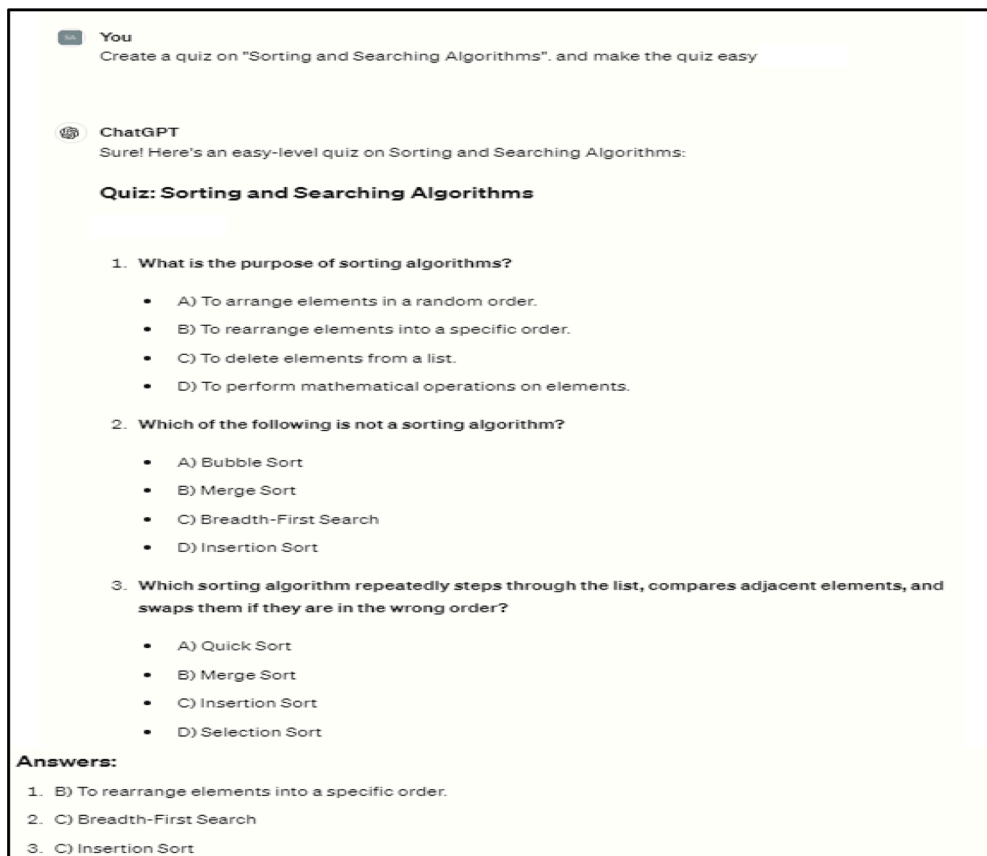


Fig. 14. Screenshot of easy quiz on "Sorting and Searching Algorithms" generated by ChatGPT.

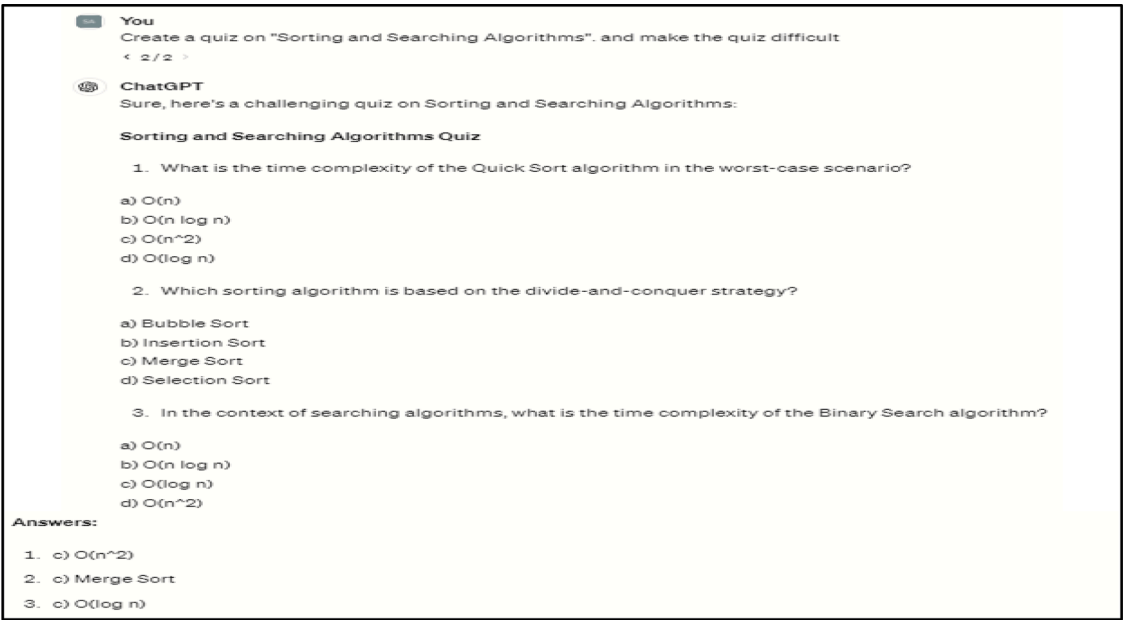


Fig. 15. Screenshot of difficult quiz on "Sorting and Searching Algorithms" generated by ChatGPT.

Project-Based Assessment Rubric					Descriptors
Criteria	Exemplary (4 points)	Proficient (3 points)	Satisfactory (2 points)	Needs Improvement (1 point)	
Research Quality	- Extensive and thorough research; demonstrates deep understanding. - Information is accurate and well-cited. - Insightful and original analysis.	- Adequate research; demonstrates good understanding. - Most information is accurate and cited. - Analysis is clear but may lack depth.	- Basic research; demonstrates limited understanding. - Some information may be inaccurate or poorly cited. - Analysis is superficial.	- Minimal research; demonstrates poor understanding. - Information is often inaccurate or missing citations. - Analysis is lacking or irrelevant.	• Research Quality: • Exemplary: Demonstrates an exceptional level of research with comprehensive understanding, accurate and well-cited information, and original analysis. • Proficient: Shows good research with a solid understanding, mostly accurate information, and clear analysis, though it may lack depth. • Satisfactory: Includes basic research with limited understanding and some inaccuracies, with superficial analysis. • Needs Improvement: Shows minimal research with poor understanding, inaccuracies, missing citations, and lacking or irrelevant analysis. • Creativity: • Exemplary: Displays a high level of originality and innovation, engaging the audience with an exceptional approach. • Proficient: Exhibits creativity with a good approach and engages the audience effectively. • Satisfactory: Shows some creativity with a somewhat original approach, but limited engagement. • Needs Improvement: Lacks creativity with a conventional approach and little audience engagement. • Presentation: • Exemplary: Highly polished and professional, with exceptional visuals or multimedia that enhance the project, and a clear, confident, and engaging delivery. • Proficient: Well-organized and professional presentation, effective visuals or multimedia, with clear and confident delivery. • Satisfactory: Organized but lacking polish, basic visuals or multimedia, with somewhat clear delivery but lacking engagement. • Needs Improvement: Disorganized and unprofessional presentation, ineffective or missing visuals or multimedia, with unclear or disengaging delivery.
Creativity	- Highly original and innovative approach. - Shows exceptional creativity in concept and execution. - Engages and captivates the audience.	- Original and creative approach. - Shows creativity in concept and execution. - Engages the audience effectively.	- Some creativity shown. - Approach is somewhat original but may be conventional. - Limited audience engagement.	- Minimal creativity. - Approach lacks originality. - Little to no audience engagement.	
Presentation	- Presentation is highly polished and professional. - Visuals and/or multimedia are exceptional and enhance the project. - Clear, confident, and engaging delivery.	- Presentation is professional and well-organized. - Visuals and/or multimedia are effective and support the project. - Delivery is clear and confident.	- Presentation is organized but may lack polish. - Visuals and/or multimedia are basic but functional. - Delivery is somewhat clear but lacks engagement.	- Presentation is disorganized and lacks professionalism. - Visuals and/or multimedia are ineffective or missing. - Delivery is unclear or disengaging.	

Fig. 16. Screenshot of rubric template generated by ChatGPT.

understanding lesson plans; but, if it is not supervised by a human, it occasionally generates generic or inappropriate responses. Moreover, Knewton, Dreambox, and Smart Sparrow are primarily utilized for differentiated instruction. They offer individualized learning paths so that every student can efficiently learn at their own speed. When vast datasets are available, however, there is an inherent need for successful adaptation, which can be challenging to curate in small classrooms. By examining student actions and learning preferences, Squirrel AI is at the leading edge of providing individualized virtual tutoring experiences; yet, it is highly dependent on the precision and efficacy of the input data [123,124]. Through cutting-edge voice recognition and live transcription, respectively, Brainia and AI-Live Captioning promote inclusivity for

access and give various learners a powerful platform for self-expression. However, the ability of technology to recognize accents and accurately transcribe in a noisy environment can further limit access [125]. Through interactive adaptive content, MATHia and Duolingo, respectively, support digital learning in language and math education. Although these are quite interesting and flexible, they typically need additional gamification tweaks to maintain learners' attention over time [126]. These platforms deliver tailored support systems, personalized learning pathways, and inclusive features embracing voice recognition and live transcription. Table 8 below provides a comprehensive overview of AI tools used in education, showcasing their practical applications in teaching preparation and assessment collected from literature.

Table 8

AI tools/platform in teaching preparation and assessment.

Aspect	AI Tools/ Platform	Description	Teaching Preparation	Assessment
Course and Lesson Plan Creation	Top Hat, Microsoft Copilot, Coursebox, Planboard, Edmodo, ChatGPT	Designing and organizing course materials and lesson plans.	✓	
Personalized Learning	Dreambox, Smart Sparrow, Knewton	Platforms that adapt learning paths to meet the diverse needs of individual students.	✓	
Auto-Grading	Gradescope	Automated system for grading assignments and exams efficiently.		✓
Pinpointing Knowledge Gaps	Exact Path	Adaptive assessments to identify areas where students are struggling and then provide personalized learning paths to improve them.		✓
Test Prep	ExamSoft, R. Test, ClassPoint, Quizlet	Resources designed to help students prepare for tests through practice and feedback.		✓
Virtual Tutoring	Zoom, Microsoft Teams, Labster	Video conferencing for virtual classes and virtual labs for remote experiments.	✓	
Student Feedback	Turnitin Feedback Studio	Tool for giving detailed feedback on student work, including originality and areas for improvement.		✓
Adaptable Universal Access	Brainia, Ai-Live Captioning	Tools for improving accessibility, including voice recognition and live captioning for diverse learners.	✓	
Digital Learning	MATHia, Duolingo, Khanmigo	Platforms that enhance digital learning experiences through interactive and adaptive content.	✓	

In the table above, the "Teaching Preparation" column indicates whether the tool is primarily used to prepare teaching materials and techniques, whereas the "Assessment" column indicates whether the tool is used to evaluate and assess student performance.

4.16. Prompt engineering for teaching preparation and assessment

It is worth mentioning that prompt engineering has emerged as a significant aspect concerning ChatGPT and other GAI, as highlighted by White et al. [39]. Scholars have investigated different methods for constructing prompts that can effectively steer the responses from the tool [127]. Prompt engineering is crucial for efficiently utilizing ChatGPT and other GAI-driven solutions in educational contexts. As AI applications in education proliferate, the necessity for accurately created prompts becomes increasingly essential, especially for tasks such as lesson preparation, assessment formulation, and instructional material development. Effective ways for prompt building encompass objective-driven prompts, progressive prompting, and role-specific prompts. Objective-driven prompts are designed to accomplish particular educational aims, such as "Create a 45-minute lesson plan on photosynthesis for high school students, incorporating objectives, interactive activities, and an assessment quiz," which directs the AI to provide a coherent and related response. Incremental prompting is deconstructing complex tasks into smaller, manageable components; for example, an educator can initiate with an outline on "climate change and its impact" and thereafter prompt for each specific topic, enhancing control over material depth and quality. Role-specific prompts direct the AI to assume a specific viewpoint or style; for instance, "Act as an interactive tutor and elucidate the Pythagorean theorem to a middle-school student, incorporating examples and questions to assess comprehension," thereby prompting the AI to engage in an interactive, conversational approach conducive to student involvement. Although prompt engineering is an advantageous resource for educators, it presents certain limitations. Ambiguity and misinterpretation may arise from ambiguous suggestions, resulting in irrelevant responses and necessitating careful refinement of language. Achieving a balance between content depth and scope can be challenging; broad prompts, such as "Explain human biology," may result in responses that are either excessively simplistic or overwhelmingly complex. Targeted suggestions on particular subtopics can facilitate more accurate responses. Moreover, there exists a potential for bias in responses, as AI-generated content may mirror the biases present in the training data [128].

In the context of leveraging ChatGPT for teaching preparation and assessment, Table 9 presents a series of illustrative prompts that educators can utilize to create and develop highly customized educational content. These prompts enable the generation of comprehensive lesson outlines, diverse instructional activities tailored to various learning styles, and assessments such as quizzes. Additionally, they assist in the creation of structured grading rubrics designed to provide clarity in evaluation and mechanisms for delivering constructive feedback that fosters student improvement and engagement. The table identifies the prompts linked to teaching preparation and assessment, providing educators with a simplified method to improve both instructional planning and evaluative processes.

While the above table primarily reflects outcomes from the reviewed literature, it also includes original synthesis and commentary provided by the authors to offer a nuanced understanding of the implications of these findings. This approach enables a more comprehensive interpretation that situates the literature within current educational frameworks and practices, providing valuable insights for both research and practical applications in education.

ChatGPT and other AI platforms can help educators with teaching preparation and assessment by automating routine activities, generating a variety of instructional materials, and instantly offering feedback about teaching strategies. By optimizing prompt construction, educators should be able to effectively leverage AI for tailored content creation, simulation of teaching scenarios, and analysis of assessment results to realize improved learning outcomes.

Expanding from these capabilities, the integration of AI tools into ERP and e-learning systems such as Canvas offer broader opportunities for streamlining educational processes and enhancing engagement, as

Table 9
Categories of educational tasks and the corresponding ChatGPT prompts.

Category	Illustrative ChatGPT prompt	Teaching Preparation	Assessment
Lesson Planning	"Generate a detailed lesson plan on the Industrial Revolution for high school students, including objectives, key concepts, discussion questions, multimedia resources, and homework assignments."	✓	
Varied Instructional Activities	"Suggest differentiated activities for a middle school science lesson on photosynthesis, catering to visual, auditory, and kinesthetic learners, with group work and individual tasks."	✓	
Quizzes and Assessments	"Create a quiz for a history unit on ancient Egypt, including a mix of multiple-choice, short answer, and matching questions that assess students' understanding of key dates, figures, and cultural practices."		✓
Structured Grading Rubrics	"Design a rubric for evaluating student presentations on environmental sustainability, with criteria for content accuracy, creativity, visual aids, and public speaking skills."		✓
Constructive Feedback Mechanisms	"Generate constructive feedback for a student essay on the effects of climate change, focusing on strengths in argumentation, areas for improvement in evidence support, and suggestions for further reading."		✓
Interactive Learning Modules	"Develop an interactive learning module on the water cycle for elementary students, incorporating animations, interactive diagrams, and quizzes that test comprehension at each stage of the cycle."	✓	
Project-Based Learning Plans	"Create a project-based learning plan for a high school economics class, where students develop a business proposal. Include project milestones, collaborative activities, research components, and presentation guidelines."	✓	
Scaffolded Learning Tasks	"Design a series of scaffolded tasks for a language arts unit on persuasive writing, starting with brainstorming sessions, followed by drafting, peer review, and final revisions."	✓	
Student Self-Assessment Tools	"Create a self-assessment checklist for students to evaluate their understanding and progress in a math unit on algebraic equations, with sections for self-reflection and goal setting."		✓

Table 9 (continued)

Category	Illustrative ChatGPT prompt	Teaching Preparation	Assessment
Culturally Responsive Content	"Generate culturally responsive teaching materials for a world history lesson on the Silk Road, ensuring diverse perspectives and including primary sources from different regions involved."	✓	
Adaptive Learning Paths	"Design an adaptive learning path for students struggling with fractions, offering additional practice problems, video tutorials, and instant feedback on common mistakes."	✓	
Collaborative Learning Activities	"Suggest collaborative activities for a social studies lesson on community building, such as group debates, role-playing scenarios, and cooperative research projects."	✓	

will be elaborated on below.

This integration has the potential to enhance the efficiency, functionality, and overall usability of both the ERP and e-learning systems. There is the possibility of using AI-powered chatbots that can reply instantly to every query, reducing the workload on staff while increasing student engagement. For example, integrating Dialogflow or Kommunicate into Canvas or Moodle can support real-time interactions and improve accessibility [129]. AI can also automate content creation and personalization, such as generating quizzes, presentations, and subtitles, as demonstrated by the "LearnStreamAI" project [130]. Predictive analytics further enable proactive decision-making, helping identify at-risk learners in e-learning systems or optimizing resource allocation in ERP systems [131]. Adaptive learning paths and collaborative environments can be created using AI, tailored to individual user needs. Moreover, cloud-based AI solutions like Google Cloud AI or AWS facilitate scalable and cost-effective integration. Ethical considerations are paramount, wherein AI systems should ensure data privacy and transparency to build trust and improve regulatory requirements such as GDPR [132]. Embed continuous feedback and evaluation mechanisms that refine the functionality of AI over time to meet organizational goals better. These steps will allow AI to transform ERP and e-learning systems into highly efficient and user-friendly platforms by following a structured implementation framework.

5. Discussion and future directions

5.1. Discussion

The findings from the first research question covered a broader research gap by systematically comparing AI-supported and traditional methods across a wide range of educational tools, disciplines, and metrics of student success. It addresses both the strengths and weaknesses of these approaches, which were only partially explored in previous studies such as [8,13,40], and provides a more well-rounded examination of their comparative effectiveness in education. The findings also reveal that AI-powered tools offer significant advantages in certain educational domains. While it concludes that keywords such as "Traditional education methods vs AI," "Balancing academic integrity," "ChatGPT across teaching preparation and assessment" and "prompt engineering," have not been thoroughly researched. Moreover, the comparison between AI-supported education methods and traditional teaching methods reveals that while AI tools offer numerous advantages,

particularly in terms of efficiency, personalization, and accessibility, they are not without drawbacks, as illustrated in Table 5. Integrating AI in education introduces ethical challenges, particularly around data security and the long-term impact on learning behaviors. Data privacy is a major concern; educational AI systems often collect extensive personal data on students, which, if exploited, can lead to breaches of confidentiality and misuse of information. Ensuring compliance with regulations such as the General Data Protection Regulation (GDPR) is essential to safeguarding student information. Beyond security, AI's influence on learning behaviors also requires careful attention. AI tools, if overused, may hinder students' critical thinking and independence, as learners may rely too heavily on automated systems for guidance and feedback. To address these concerns, educators and policymakers are urged to develop frameworks that encourage responsible AI use, promoting a balanced approach where technology supports—rather than replaces—student-driven learning and cognitive development.

The analysis in this study demonstrated that the use of ChatGPT in academic writing and literature reviews presents both opportunities and challenges. Its strengths lie in enhancing productivity, improving language clarity, and streamlining the review process, as shown in Table 6. In academic writing, ChatGPT has proven to be an invaluable tool for drafting and refining text, generating citations, and overcoming writer's block. Its capabilities extend to literature reviews, where it excels in extracting relevant literature, summarizing key findings, and identifying emerging trends. This multifaceted support significantly enhances the efficiency and quality of the review process, enabling researchers to produce more comprehensive and insightful analyses. However, the observations from this research proposed that human involvement remains essential to validate AI-generated content and maintain scholarly standards. As such, ChatGPT should be seen as a complementary tool rather than a replacement for traditional academic writing and literature review practices.

The author believes that prohibiting the utilization of AI technology, such as ChatGPT, is unwise. While several schools may restrict the use of ChatGPT throughout the K-12 phase [133,134], such limitations are justifiable given that students are at a pivotal point in their cognitive and skill development. Higher education students should utilize available resources and technologies to accomplish work effectively and enhance their growth chances. If ChatGPT can flawlessly do homework given by university teachers, they should reevaluate the structure of these assignments to determine if they could significantly enhance student learning. As these students reach the workforce, they may utilize ChatGPT technology to do numerous duties with precision. This prompts an inquiry into the necessity of human involvement in certain jobs or the potential for their replacement by robots. Consequently, higher education students require distinct skills and competencies beyond those of ChatGPT and must persist in their self-development to get a competitive edge in the employment market. With technological advancements, the AI revolution has started. Existing policies and human oversight are inadequate to address this transformation; so, robust mechanisms must be implemented to govern the utilization of AI technology. Consequently, these technologies should not be indiscriminately banned but rather carefully controlled to foster the advancement and utilization of AI technology.

The findings of this study suggested that the integration of AI tools like ChatGPT into global educational institutions presents a complex landscape of opportunities and risks. As highlighted in Table 7 and Fig. 11, the diverse strategies adopted by institutions worldwide reflect a balance between fostering the innovative use of AI and maintaining academic integrity. While some institutions like Sciences Po and New York City Public Schools have implemented strict bans, others, such as Imperial College London and the University of Sydney, encourage the responsible and transparent use of AI. These varying approaches reveal global differences in how educational institutions view the potential of AI to either enhance learning or pose risks such as cheating and undermining critical thinking skills. The challenge remains for

institutions to design policies that promote the ethical use of AI while protecting the integrity of academic work.

While the previous studies provide valuable insights into the role of ChatGPT in education, they do not comprehensively explore the potential for AI tools to enhance educators' professional development in the specific areas of teaching preparation and assessment. Furthermore, none of the studies address the best practices for constructing effective AI prompts, which are crucial for maximizing the efficiency and impact of these tools. This study's outcomes implied that the use of ChatGPT in lesson planning helps educators move beyond traditional textbook methods, incorporating interactive elements that enhance student engagement and learning outcomes. Additionally, Figs. 13, 14, 15, and 16 proved the capabilities of ChatGPT to assist in creating detailed grading rubrics, creating quizzes of varying difficulty levels, and creating assessment materials. While ChatGPT offers versatile applications across teaching preparation and assessment, other AI tools provide specialized functionalities as indicated in Table 8. For example, tools like Top Hat and Microsoft Copilot are focused on course and lesson plan creation, while platforms such as Gradescope and Exact Path are geared towards auto-grading and pinpointing knowledge gaps. The data from this investigation revealed that each tool has its strengths, and educators may benefit from using a combination of AI tools to address various aspects of teaching and assessment.

Finally, the evidence gathered in this research pointed to the fact that prompt engineering is crucial for optimizing the efficacy of ChatGPT in educational preparation and evaluation. Through the creation of targeted prompts, as referenced in Table 8 instructors may produce tailored learning resources that address the unique requirements of their students, including differentiated training for varying proficiency levels.

5.2. Future directions

As depicted in Fig. 17, this study suggests four critical avenues for future research that promise to enhance the efficacy and integration of AI in educational settings:

- **Ethical and Social Considerations:** Addressing ethical challenges is paramount in the adoption of AI technologies. Future research should focus on the development of robust frameworks that ensure data privacy, combat biases, and promote social equity. It is essential to establish clear guidelines that facilitate the responsible use of AI tools, guaranteeing that these technologies are accessible and beneficial to all students, irrespective of their socio-economic backgrounds.
- **Integration of Cognitive Science:** Merging AI with cognitive science can significantly improve how these technologies support learning. Research should be directed towards creating AI tools that are sensitive to the cognitive processes of learners, thereby enhancing the personalization of educational experiences. This integration can help tailor learning environments that respond adaptively to the needs of individual students, making education more effective and engaging.
- **Long-Term Impact on Student Learning:** There is a compelling need to explore the long-term consequences of AI on educational outcomes. Future studies should investigate how sustained interactions with AI tools like ChatGPT affect critical thinking, knowledge retention, and student creativity. Understanding these long-term impacts will provide invaluable insights into how AI can be optimally utilized to support lifelong learning and intellectual growth.
- **Integration of Personalized Prompts:** The potential of AI to generate personalized learning prompts represents a significant opportunity to revolutionize educational engagement and effectiveness. Future research should concentrate on refining the algorithms behind prompt generation to ensure that they are highly relevant and contextually appropriate for each student. This approach will not

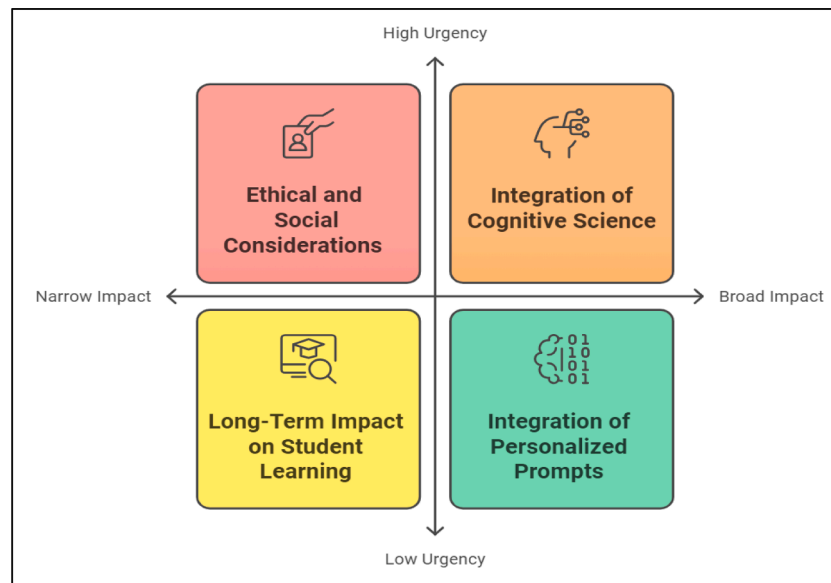


Fig. 17. Future research directions of AI in education based on urgency and impact.

only increase student engagement but also enhance the overall learning process by providing customized support that aligns with individual learning trajectories.

Future research should attempt to build an educational ecosystem where AI tools complement rather than replace traditional teaching, used ethically, fairly, and to improve learning. By tackling these study areas, educators and policymakers may create a future where AI technology is a transformative force that adapts to students' changing needs, respects ethics, and promotes holistic educational goals. Fig. 17 shows each quadrant reflects different strategic priorities, helping institutions and researchers prioritize their focus areas based on the urgency and reach of impact in educational contexts. This future-oriented agenda aims to comprehensively leverage the capabilities of AI tools in education, ensuring that these technologies are used to complement traditional educational methods rather than replace them. By addressing these targeted areas, future research can help realize the full potential of AI in enhancing educational practices and outcomes.

6. Limitations and conclusion

6.1. Limitations

This study is limited by various aspects pertaining to selection and scope. The literature analysis solely examines papers from December 2022 to October 2024, a timeframe selected to encompass the latest advancements subsequent to ChatGPT's introduction. This era may omit emerging studies that could provide further insights. Language constituted a limiting constraint, as only English-language academic publications, reviews, and conference papers were incorporated, potentially excluding pertinent research published in other languages or informal formats, such as blogs and preprints. Moreover, although the study offers a comprehensive viewpoint, the intrinsic intricacy of classifying and contrasting the educational effects of AI and conventional teaching techniques incorporates subjectivity into the evaluation. This review primarily examines the applications and limits of ChatGPT in higher education, which may restrict the generalizability of findings to K-12 education.

6.2. Conclusion

In conclusion, the current study systematically explores the role of

ChatGPT and other AI tools in education by discussing their comparative effectiveness, institutional adoption, and professional development applications. The findings emphasize the unique advantages of AI-supported methods over traditional teaching, especially regarding enhancement in students' performance and academic practices, while also pointing out that traditional methods remain indispensable. This work investigates the capabilities of ChatGPT related to academic writing, literature reviews, and translation, revealing its potential as a very useful tool for both students and educators. The paper further discusses institutional strategies for achieving a balance between innovation and academic integrity and provides insights into policy frameworks that foster responsible AI use. It finally outlines best practices in constructing prompts that optimize ChatGPT's effectiveness for lesson planning, assessments, and personalized feedback, outlining the transformative potential of ChatGPT in education. These findings contribute to a holistic understanding of AI's impact on education and provide actionable recommendations on leveraging its capabilities responsibly.

CRediT authorship contribution statement

Sayeed Salih: Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Omayma Husain:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Mosab Hamdan:** Writing – original draft, Supervision, Funding acquisition. **Samah Abdelsalam:** Writing – review & editing, Writing – original draft. **Hashim Elshafie:** Resources, Funding acquisition. **Abdelwahed Motwakel:** Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Table 10

Table 10

Analysis of the included studies.

No.	Ref	Methodology	Publication Year	Key finding	Category
1	[40]	survey and post-test assessment	2024	The survey explores students' preference for traditional and AI-driven problem-solving methods, finding that while AI can clarify complex ideas, in-person instruction remains preferred, suggesting a balanced learning strategy.	Comparative effectiveness of AI-supported education versus traditional methods
2	[41]	Comparative approach	2024	The research underscores the essential nature of integrating ChatGPT into learning by providing interactive and personalized support, helping students grasp and explore complex concepts more effectively compared to the traditional education approach.	
3	[39]	Systematic literature review	2024	Explores the integration of ChatGPT in education, highlighting that ChatGPT can be efficiently integrated into education to automate tasks, improve student learning, and increase productivity, unlike traditional education methods.	
4	[42]	Meta-analytic approach	2024	The study offers a theoretical foundation and empirical support for the scientific utilization of Gen-AI and the formulation of educational technology policy.	
5	[47]	Systematic review	2023	Systematically reviews the educational applications of ChatGPT and other AI tools, providing insights into the educational applications of ChatGPT for future research.	
6	[8]	Exploratory study	2023	Explores the transformative potential of ChatGPT vs traditional tools in revolutionizing education, suggesting avenues for future research. The findings indicate that AI may significantly assist kids with learning impairments in realizing their full potential.	
7	[13]	Conceptual analysis	2023	Explores opportunities for improving the education system using ChatGPT. The study reveals that ChatGPT serves as a valuable resource compared to traditional tools for language instruction due to its ability to translate text between languages.	
8	[114]	qualitative methodology and document analysis	2023	The results indicate that GAI models like ChatGPT can offer targeted resources and support systems, including lesson plans, to educators and prospective educators. It also emphasizes the necessity for more study to investigate optimal strategies for using ChatGPT in lesson design.	Professional development of educators in teaching preparation and assessment
9	[115]	Mixed-methods approach	2023	Explores the various challenges and opportunities presented by ChatGPT in teacher education. The article discussed the several ways ChatGPT can be misused, including writing content that an author uses as their own (e.g., plagiarism and ghostwriting).	
10	[113]	Qualitative survey	2024	Investigates how teachers view ChatGPT as a tool for education, including benefits, challenges, and the impact on their teaching practices.	
11	[70]	Interpretative phenomenological analysis (IPA)	2024	The analysis revealed four themes under motivating factors that encourage teachers to adopt ChatGPT for their educational purpose. These include Theme 1: Exploration of innovative education technologies, Theme 2: Personalization teaching and learning, Theme 3: Time-saving and Theme 4: Professional development.	
12	[117]	Literature review	2023	Discusses the potential benefits of ChatGPT in promoting teaching and learning. The finding recommended policymakers, researchers, educators, and technology experts to explore constructive use of GAI technologies to enhance education and assist student learning.	
13	[118]	Literature review	2023	findings underscore the benefits and challenges of ChatGPT in higher education, highlighting its potential for adaptive learning, personalized feedback, research support, automated administrative services, and innovative assessments, while also addressing academic integrity concerns and limitations.	
14	[116]	Exploratory study	2023	This study examines the role of instructors in AI-based evaluation and the problems of adopting AI-powered technologies in education. The study concludes with ideas on how to overcome difficulties and improve AI in educational assessment.	

(continued on next page)

Table 10 (continued)

No.	Ref	Methodology	Publication Year	Key finding	Category
15	[130]	Case study	2024	Integration of AI-driven chatbots in Moodle LMS enhances student interaction, feedback, and material preparation.	The balance between the opportunities and risks of AI tools in education institutions.
16	[132]	Qualitative and Quantitative	2023	Integration steps and examples of AI technologies, particularly GPT-3, into learning management systems.	
17	[129]	Case study	2022	AI chatbot in Moodle LMS effectively reduces repetitive queries and supports student engagement.	
18	[131]	Quantitative	2024	AI-driven platforms significantly influence student satisfaction and intentions to use e-learning platforms.	
19	[96]	Case study	2024	Identifies factors influencing students' attitudes toward ChatGPT based on the technology acceptance model. The recommended model was empirically tested with partial least squares structural equation modelling using 375 student survey responses.	
20	[95]	document analysis	2023	Through a review of pertinent literature, the study has looked at the opportunities, difficulties, and approaches related to the use of ChatGPT in higher education.	ChatGPT in academic writing and literature reviews
21	[43]	Analytical review	2024	The study discussed academic dishonesty detection and prevention challenges and offered university ideas for the ethical and responsible use of these technologies. The author suggested policies and processes, provided training and assistance, and used different approaches to identify and prevent cheating.	
22	[135]	Empirical analysis	2024	Investigate the varying effectiveness of ChatGPT in different subject areas (coding, finance, math, and general public queries) and its potential issues such as generating inaccurate data and plagiarism. The findings suggested approaches to balance the opportunities and risks of AI tools in higher education.	
23	[93]	Thematic review	2024	Findings shed light on the prospects and challenges of using ChatGPT for teaching, learning, and research, focusing on practical applications and potential drawbacks.	
24	[103]	Systematic Literature Review	2023	The study concluded that ChatGPT has the potential to improve academic and librarian-related activities; nonetheless, it is essential to reevaluate the ethical implications of utilizing such technology.	
25	[9]	Ethnographic research	2023	The paper examines the possible advantages and drawbacks of ChatGPT, along with ways for mitigating the observed issues. It underscores the pressing necessity for explicit regulations, norms, and frameworks to judiciously incorporate ChatGPT into higher education.	
26	[17]	Ethical review	2023	The finding suggested that ChatGPT can enhance critical thinking and cognitive flexibility, supporting diverse learning styles and closing educational gaps. By leveraging AI tools, educators can foster character development and positive psychological outcomes in students.	
27	[94]	online questionnaire	2023	Analyzed university students' perceptions of GAI technologies in higher education. A survey of 399 students in Hong Kong revealed a largely positive attitude towards GAI, +and participants recognized the potential for personalized learning assistance, support in writing and brainstorming, and enhanced analytical skills.	
28	[91]	Qualitative review	2024	Findings imply that colleges may manage academic integrity challenges by being proactive and ethical. Academic dishonesty can be detected by developing comprehensive policies, offering training and support, and implementing measures.	
29	[109]	Literature review of AI misuse cases in academic settings	2023	Identifies AI's impact on academic integrity, emphasizing the need for strict guidelines to prevent plagiarism.	
30	[43]	Case study analysis and policy review in educational tech	2024	Highlights ChatGPT's role in potential cheating; recommends policy frameworks to ensure ethical AI use.	
31	[111]	Systematic review analyzing studies on AI dialogue systems and their effects on students' cognitive abilities	2024	Over-reliance on AI dialogue systems can negatively impact students' critical thinking and problem-solving skills, highlighting the need for balanced AI integration in educational settings.	
32	[136]	Examination of plagiarism cases involving AI	2023	Assesses how AI may facilitate plagiarism; calls for increased awareness and citation standards for AI-generated content.	
33	[81]	Case studies and surveys	2024	Results highlight how AI tools like ChatGPT can enhance productivity in academic writing and research, and discuss their benefits and limitations.	
34	[72]	Review and case studies	2023	Outcomes accentuate that ChatGPT may produce misleading or incorrect data, necessitating careful verification by researchers. It recommends the academic and scientific community formulate essential standards for the acceptable utilization of LLMs, particularly ChatGPT, in research and literature reviews.	

(continued on next page)

Table 10 (continued)

No.	Ref	Methodology	Publication Year	Key finding	Category
35	[75]	Comparative analysis	2024	Examines how ChatGPT rates essays compared to human raters, focusing on the consistency and accuracy of ChatGPT as a powerful tool in academic writing and literature review.	
36	[71]	Analytical review	2024	Research examines the role of ChatGPT in academic writing, including its positive and negative impact on research practices and writing quality.	
37	[77]	systematic review	2023	Systematically reviews the impact of ChatGPT implementation in education, offering implications for educational practices in academic writing and literature review.	
38	[74]	qualitative and semi-structured interviews	2024	Students who use Chat GPT for academic writing were interviewed through a qualitative methodology including semi-structured interviews. The research indicated that Chat GPT offers users information about its origins, benefits, and reliability.	
39	[89]	Review	2023	Provides a taxonomy of hallucinations in large language models (LLMs) and identifies factors contributing to hallucinations. Discusses detection methods, mitigation strategies, and limitations of retrieval-augmented LLMs, with future research directions.	
40	[90]	qualitative and semi-structured interviews	2024	Identifies four risks of ChatGPT in academia: plagiarism, overdependence on technology, reduced critical thinking, and diminished assignment quality. Proposes mitigation strategies, including plagiarism detection, clear usage guidelines, and awareness campaigns.	

Data availability

No data was used for the research described in the article.

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